StockSmart

End-to-End Stockroom Management System

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ES96: Engineering Design Project

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0 - Executive summary

The Spring 2026 ES96 class at Harvard University designed and built an integrated billing system for shared stockrooms at the Wyss Institute. The class first spent five weeks researching the client, through online research and discussions with various stakeholders at the Wyss: researchers, department heads and operational staff.

After building an understanding of the client, the class narrowed their focus on identifying potential problems to be solved through an engineering design process. Project team members conducted further interviews and group brainstorming sessions generated lists of ideas. Further class discussions and back-and-forth with the client led the team to a problem statement regarding shared stockrooms' per-visit billing practices and sub-optimal inventory management practices, which were causing financial inequity between labs and operational downtime.

Having selected a problem to work on, the class began brainstorming solutions. Through a strategic process, solution ideas were combined or eliminated until one remained: an app on stockroom-specific phones to enable at-shelf scanning of items. The team formed groups to work on the frontend app, backend website, and phone charging station and began prototyping the solution. After several rounds of revisions with frequent communication to ensure alignment between teams, the product was presented to the client.

Implementing this solution would involve incorporating client feedback to produce a deployment-ready version of the system and producing final versions of hardware and software before initiating final installation and turnover to the client.

1 - Client Overview

1.1 - Introduction

In this class, our team was tasked with researching, identifying a problem, and developing a solution for a client. This semester the client we worked with was the Wyss institute at Harvard University. Under current management practices at the Wyss Institute, labs are billed on a per-visit basis which is then averaged across all supply room access swipes, meaning swipes in which a single inexpensive item was taken are billed the same amount as swipes in which several expensive items were taken. Furthermore, the lack of an efficient stock management system means researchers have to swipe into several stock rooms to find an item that may not be in stock anywhere. The result is financial inequity and operational inefficiencies that quietly erode the research capacity of labs across the Institute.

The solution that the team came up with is a two-part stockroom management system comprising a mobile app and an administrative website. The stock rooms' functionality is built around an NFC tap-to-login, mobile checkout that allows for administrative tracking of stock room usage and more accurate per-lab billing. This will solve the financial inequity while creating a seamless transaction system that allows for fully automated inventory tracking. This report documents the full arc of the work that led to our solution, starting with client assessment and problem identification all the way to prototyping and final delivery to the client.

1.2 - Client Overview

The Wyss Institute is a cross-disciplinary, biology-inspired scientific research institute at Harvard University. Founded in 2009 with a $125 million philanthropic gift from Hansjorg Wyss, its core mission is to solve one of the complex issues facing Harvard research: the carryover into industry. Patent legalities make it very difficult for researchers at Harvard to seamlessly transfer over to private and public sectors since all conduct is under the umbrella organization of Harvard University. This prevents clinical trials or further development from being done once research is complete due to patent ownership. The Wyss Institute, falling organizationally under Harvard, connects the necessary resources such as funding, direct connections to industry, as well as direct connections with leading clinical medical research laboratories.

The institute pioneered a reverse approach in bioengineering where researchers draw inspiration from biological and natural design principles to solve pressing challenges in healthcare and the environment. The operational model of the Wyss is what sets it apart from other research institutions and incubator startup organizations. It combines the depth of academic science research with the flexibility of private industry to make deliverable technologies and medicines that are market-ready. Central to this model is the Institute's Advanced Technology Team which is composed of 40 staff members. The team is comprised of both scientists and engineers each with years of prior experience in industry and product development. They are principally tasked with directly integrating research teams and helping with the research to industry pipeline. This team works alongside business development professionals, IP specialists, and entrepreneurs to give the institute an internal infrastructure that most universities lack entirely. Faculty, staff, fellows, and students in the Wyss are organized into five technology focus areas, each of which is headed by core faculty with world-leading expertise in their respective fields. Additionally, the Wyss Institute provides pivotal research spaces that allow for cross pollination and collaboration between labs and research projects.

The institute is located across two primary campuses – the Fenway campus and the Longwood medical campus. Additionally, each researcher works at their own respective lab and regularly travels between the Wyss Institute and Harvard in order to complete their research. Furthermore, researchers are drawn not only from Harvard, but also from twelve collaborating academic institutions and hospital partners including Brigham and Women's Hospital, Boston Children's, Dana Farber, and Mass General. This network of collaborators gives Wyss researchers direct access to clinical insight and patients and allows for the embedding of real-world medical needs into the early stages of the design process.

The substantial results of the Wyss speak for themselves. Since its founding, the institute has accumulated more than 1,750 issued patents, 155 licensing agreements, and contributed to the launching of 71 startups with many more in the works. Notable projects of the past include Emulate Inc., which commercialized the Institute's organ-on-a-chip technology for in vitro drug development and disease modeling; Dyno Therapeutics, which raised a $100 million Series A to develop enhanced gene therapy delivery vehicles in partnership with Novartis, Sarepta, and Roche; and ReadCoor, which was a startup founded at the Wyss that was sold for $350 million. Across therapeutics, diagnostics, soft robotics, and sustainable materials, the institute has built a tremendous track record of moving ideas from the bench to market at a pace that is rare for academic research.

Financially the institute operates as a non-profit under the Harvard University umbrella, funding its work through a combination of philanthropic gifts, federal grants from agencies such as the NIH and DARPA, and revenue from licensing agreements and industry partnerships. This structure gives the Wyss flexibility to pursue high-risk research that neither traditional academic labs nor private companies would typically take on. An example of this is the Women's Health division at the Wyss. Women's health is a historically under-researched and under-funded field in academics. The flexible financial structure in conjunction with the Wyss mission statement allows for it to not only execute this type of research, but let it be a main focus area.

Taken together, the Wyss institute represents a genuinely novel institutional model that is unprecedented in its success. The Wyss combines the strengths of incubator startups, academia, and industry implementation. This understanding was pivotal to the ES-96 team, so that we could understand what the core identity of the Wyss Institute is and how our expertise could best assist in carrying out its mission. With this in mind, our team spent significant time researching each focus area: Therapeutics, Women's Health, Sustainability, Medical Technology, and Diagnostics.

1.3 - Research and Client Interviews

1.3.1 - Therapeutics Division

Introduction to Therapeutics

The Therapeutics division of the Wyss Institute operates at one of the most ambitious frontiers in modern biomedical research. Rather than focusing just on conventional drug discovery, the Wyss takes a multifaceted strategy to advance the development of more effective therapeutics, spanning approaches like predictive bioanalytics, machine learning, synthetic biology, mRNA therapies, small molecules, and even new preclinical models like human organs-on-chips. What makes this branch unique is its structural fusion of disciplines like materials science, immunology, and engineering, allowing researchers to draw inspiration from unorthodox sources and truly explore the institute's motto of "biology inspired engineering." The result of this approach and the structural focus of commercialization are entire therapeutic platforms that are scalable, cross-functional systems designed to address or model classes of diseases, targets, and patients.

To fully leverage this interdisciplinary approach and commercialization pipeline, the Therapeutics branch partially benefits from direct collaborations with Harvard Medical School, affiliated hospitals, and a wide network of industry partners across Cambridge and Boston. This positioning strategically accelerates pre-clinical validation and gives researchers more direct paths towards clinical relevance, impact, and partnerships. In particular, the Wyss' Validation Projects function as effective de-risking platforms that prepare technologies for commercial projects, and functions as a start-up incubator, having launched companies such as Colossal

Therapeutics ($10.2 billion valuation)[1], Gameto ($250 million valuation)[2], GC therapeutics ($65 million Series A round).[3]

Key Projects and Focus Areas

The current Therapeutics portfolio spans numerous cutting-edge focus areas, each reflecting the institutional commitment to building platform technologies rather than one-off solutions. Immunology and cell engineering sit at this department's core, with projects focused on patient-derived T-cell engineering, CAR-T therapies, and tunable hydrogels for immune cell population manipulation. DNA nanotechnology represents another major pillar for the institute, where the Wyss Institute has developed a suite of diverse, multifunctional DNA nanotechnological tools engineering to overcome specific bottlenecks in the development of novel therapies, such as DNA nanoswitches and a "Lab-on-a-Molecule."

Furthermore, targeted drug delivery exploration within the institute is championed by well developed platforms, such as AminoX, using non-standard amino acids engineered into protein drugs that only become active within the tumor microenvironment, and NanoX, drug targeting activated locally by mechanical forces. On the other hand, synthetic biology and biomolecule manufacturing round out the portfolio and focus on quickly and easily producing a wide range of vaccines, antimicrobial peptides, and antibody conjugates, through platforms like CircaVent and CogniXense.

[1]Bloom, David. "Colossal Secures $200M to Accelerate De-Extinction and Genomic Innovation." Colossal Biosciences, January 16, 2025. https://colossal.com/colossal-secures-200m-to-accelerate-de-extinction-and-genomic-innovation/.

[2]Gameto. "Gameto's $44M Series C to Help Redefine Reproductive Care, as First Patients Recruited for Pivotal Stem Cell Phase 3 Trial." PR Newswire, August 12, 2025. https://www.prnewswire.com/news-releases/gametos-44m-series-c-to-help-redefine-reproductive-care-as-first-patients-recruited-for-pivotal-stem-cell-phase-3-trial-302527828.html.

[3]PitchBook. "Company Profile." Accessed May 1, 2026. https://pitchbook.com/profiles/company/460074-97.

SWOT Analysis

On the strengths side, their robust network of high impact scientists within their fields allows for interdisciplinary collaboration, leading to cutting-edge research, increasing grant acquisitions, and a quick response to global needs, as was seen with COVID-19 therapeutics. Unlike traditional pharma or even most academic biomedical centers, the Wyss builds therapeutic systems, like immune engineering frameworks, delivery architecture, and organ-on-chip validation tools, that generate value across multiple disease areas simultaneously. Combined with their proven commercialization pipeline, this platform effectively bridges academic research and industry, turning discoveries into marketable, patent-protected, and licensed products. As previously mentioned, this makes for unique metrics of success within their commercialization pipeline such as startup count (71), licensing agreements (155), and filed patents (4,702).

Weaknesses, however, stem directly from those strengths. The very complexity of Wyss therapeutic technologies make them costly and slow to manufacture and establish before becoming profitable. The breadth of the portfolio results in scalable and significant breakthroughs only if done fast enough to be profitable before Big Pharma or other startups with greater funding, larger personnels, or specific focuses can commercialize that technology before the Wyss. The Wyss' broad scope, despite allowing for greater interdisciplinary research among fields, can dilute their focus and optimization in a particular field, increasing their product manufacturing, equipment and personnel costs, along with introducing institutional questions of prioritization of maximal overlapping areas with greater commercialization potential versus those of smaller overlap.

As mentioned, a key success metric for the Wyss is the effectiveness of their commercialization platform, hence continuously optimizing this pipeline to include easier collaborations between groups and a clear framework post Validation project stage is an incredible opportunity to increase Wyss revenue through industry partnerships and royalties, securing greater financial stability. Entrepreneurship mentorship is already being practiced in the Wyss, as seen through their successful Lumineer fellowship cohorts spinouts. Scaling this initiative to increase the likelihood of successful ventures could also be a long-term financial stability tactic. On a macro scale, the broader shift of the biotech field toward precision and personalized medicine, combining AI with targeted drug discovery, aligns perfectly with what the Wyss builds and its core faculty.

External funding is an imminent threat, especially with current federal higher education funding cuts. This has a downstream effect of making the Wyss a less trustworthy "derisker" of biotech ventures and consequently reduces industry-academia partnerships. On the other hand, many technologies require manufacturing upscaling, which is capital intensive, resulting in a freeze of a project until funding is secured. The Wyss' commercialization focus places it in a direct and uphill competition with many Big Pharma ventures. Their greater commercialization resources and capital can outpace Wyss innovations if de-risking is complete and the start-up has spun out of the institution. On the other hand, with the capital scarce and fragile biotech ecosystem, many of these companies will not spin out if not backed by Big Pharma interest.

Interview Insights

Having completed our SWOT analysis of the Therapeutics division, we conducted an interview with Kwasi Adu-Berchie, Ph.D., a Wyss Senior Scientist, to understand his role and view of the institute. In this interview, our key takeaways were that the most pressing issues faced by the Therapeutics department were highly technical and related to their project-specific goals. However, overall logistical and structural issues also did become clear and were further explored during the interview. The poorly integrated physical location of the institute results in researchers needing to travel across the Longwood campus to process types of samples, conduct mousework, or use specific equipment the institution does not provide. This is time-intensive and requires significant planning across all labs, proving to be a major source of inconvenience and inefficiency. Furthermore, temperature, humidity, and different levels of motion can all affect sample transportation on the Longwood M2 shuttle, leading to suboptimal experiment results. Additionally, less defined pathways and strategies to market post Validation project incubation also prove tricky, as researchers are now dealing with a semi-ready project for commercialization, yet numerous tracks (licensing, spinning out into a startup, or conducting clinical trials within neighboring hospitals) to choose from and limited guidance.

1.3.2 - Sustainability Division

Introduction to Sustainability

The Sustainability division of the Wyss Institute applies principles derived from natural systems to develop transformative technologies that address escalating environmental challenges. By emulating biological processes, the division implements its innovation model to generate disruptive solutions focused on decarbonizing industries, restoring biodiversity, mitigating climate change, and enhancing planetary habitability for future generations. A key strength of the division is its conceptualization of sustainability as a health issue, establishing a direct connection between planetary and human health. This approach enhances funding prospects, increases policy relevance, and improves the potential for public adoption.

Key Projects and Focus Areas

The division maintains a diverse and well-defined technology portfolio, including cooling systems, water purification, air quality monitoring, PFAS detection, and sustainable materials.[4] There is a strong focus on developing scalable and manufacturable alternatives to harmful substances such as PFAS and plastics, supported by specialized laboratories and strategic alliances dedicated to materials innovation. Many technologies are engineered for deployment in low-resource, real-world environments rather than being limited to laboratory use. Furthermore, the adoption of novel bio-inspired methodologies results in differentiated solutions that are challenging for competitors to replicate.

The broad scope of the division's sustainability initiatives, which span multiple industries, regulatory frameworks, and markets, can dilute strategic focus and strain available resources, although this breadth may also create additional funding opportunities.[4] Technologies in sectors such as construction, cooling, and biological remediation often experience slow adoption due to conservative building codes, public skepticism, and complex regulatory environments. Additionally, performance degradation outside controlled laboratory conditions remains a significant risk for air, water, and cooling systems.

SWOT Analysis

Significant opportunities exist, including growing demand for PFAS alternatives, environmental monitoring technologies, and cooling solutions driven by rising global temperatures and refrigerant regulations. Framing sustainability technologies as health interventions opens new entry points into healthcare partnerships and funding streams. However, the division operates in a crowded, competitive market, faces regulatory hurdles and liability risks, and must navigate political uncertainty and funding instability. Scaling, manufacturing durability, and long-term maintenance remain major challenges to achieving widespread impact.

Interview

To receive more information, an interview with Dr. Emily Stoler was conducted. Dr. Stoler is a chemist by training who has spent most of her career in product development, including early work at the Warner Babcock Institute and as a high-level consultant for organizations like Nike, Coca-Cola, and DARPA. She later transitioned into drug delivery at the Wyss Institute, left to work at Ginkgo Bioworks, and returned to the Wyss three years ago, motivated by its potential to build impactful tools that had not yet fully materialized.

At the Wyss, Stoler described an incredibly collaborative and resource-rich environment, though sustainability has historically remained secondary to medtech and diagnostics. She emphasized two key priorities for projects: achieving real market viability and being realistic about scale-up. Successful projects that scale up must consider policy, economics, market demand, and manufacturing feasibility. The Wyss operates in a highly tech-driven and needs-based manner without a formal prioritization system, often serving as a connector between researchers and engineers. The institute strongly favors projects that can lead to licensing or spinouts that are supported by rigorous vetting and team-building processes.

Stoler highlights both the opportunities and challenges within the Wyss's sustainability work. The bioinspired engineering that sustainability provides is a major strength as they can leverage nature's low-energy and low-contamination solutions. This still has not made it easier to translate these ideas into deployable technologies. Regulatory complexity, fragmented supply chains, and disappearing federal funding all pose barriers that push the Wyss to reframe sustainability work around human health and seek philanthropic support. Additionally, operational inefficiencies and a lack of systems for tracking resource use reflect broader challenges in scaling and implementation. Despite all these complex issues, Dr. Stoler remains optimistic, noting that impactful sustainability solutions often emerge gradually and require long-term development to become widely adopted.

1.3.3 - Medical Technology Division

Introduction to Medical Technology

The Wyss's MedTech region involves the application of engineering principles and scientific discovery for the detection, treatment, and prevention of disease. From a Band-Aid strip, to an organ grown in a lab dish, to a soft robotic exosuit that retrains a stroke survivor's arm, the field has grown faster than almost any other sector over the past two decades. What makes MedTech so profound is how tangibly problems translate into opportunity. A patient bleeds internally after surgery. That gap in care drives researchers to invent a tough gel adhesive that bonds to wet tissue. A child is born premature with compromised neural development. That reality pushes a team to prototype a wearable mitt that tracks neuromotor and cognitive signals in infants who cannot speak for themselves.[4] The field is ruthlessly practical, and perhaps that is what draws so many engineers to it. There is nothing abstract about saving a life. The global MedTech market was valued at over $500 billion in recent years, and projections suggest continued expansion well into the 2030s, driven by aging populations, rising chronic disease burdens, and a healthcare infrastructure desperate for innovation. Yet market size alone is a misleading measure of the field's vitality. The real action happens in research institutes, university labs, and hybrid spaces that sit somewhere between pure science and product development. Nowhere in the world is that in-between space more productively occupied than at the Harvard Wyss Institute for Biologically Inspired Engineering.

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[4] Wyss Institute for Biologically Inspired Engineering. (n.d.). *Wyss Institute*. Harvard University. https://wyss.harvard.edu/

Key Projects and Focus Areas

The Wyss Institute's current research spans wearable and implantable devices, soft robotics, regenerative medicine, diagnostic platforms, surgical tools, and advanced manufacturing techniques.[5] These categories do not sit in neat, separate silos; they bleed into each other constantly. On the wearable side, the institute has developed MyoExo, a strain-sensor-based diagnostic worn on the body that detects muscle rigidity associated with Parkinson's disease and other neurological disorders. As the Wyss website describes, data obtained from patients continuously could improve monitoring of treatments and therapeutic interventions (Wyss Institute, 2025). It sounds simple enough as a concept, but the engineering challenges involved in making a sensor that is accurate, comfortable, and clinically meaningful are genuinely nontrivial. Soft robotics is another major pillar. Devices like soft robotic shoulder supports for stroke rehabilitation and flexible robots designed for endoscopic gastrointestinal procedures push the limits of what compliant, body-safe materials can do inside and alongside the human form.[6] The institute's FOAMs technology, which develops soft artificial muscles capable of smooth and efficient movement, represents years of accumulated insight into how biology engineers motion, and how humans might replicate it.[7] Then there is regenerative medicine. ReConstruct, one of the more headline-grabbing current projects, is a platform for growing, vascularizing, and implanting patient-derived tissues to enable safer breast reconstruction following cancer surgery.[8] Injectable hydrogel adhesives for muscle regeneration,

[5] Wyss Institute for Biologically Inspired Engineering. (n.d.). *Wyss Institute*. Harvard University. https://wyss.harvard.edu/
[6] Ibid.
[7] Ibid.
[8] Ibid.

soft conductive hydrogels for brain implants, and focused rotary jet spinning for heart valve fabrication round out a portfolio that is, quite literally, attempting to rebuild the human body piece by piece.[9] Manufacturing and diagnostics round out the picture. Pop-Up MEMS, an origami-inspired micromanufacturing technique, enables the production of complex three-dimensional microelectromechanical systems that traditional silicon wafer processing cannot produce at the mesoscale.[10] On the diagnostic end, the Blood Clot Dx platform represents an ultra-sensitive triage tool that can identify patients at risk for clotting events before they occur.[11] That kind of proactive, anticipatory diagnostics is where the field is heading, and Wyss is firmly out in front.

The Wyss operates under what might be called a 'Foundry Model,' a term that captures how the institute industrializes the gap between scientific discovery and commercial deployment. Rather than publishing findings and moving on, Wyss maintains roughly 40 staff engineers and business development professionals with hands-on FDA and industry experience, enabling inhouse prototyping and near-Clinical Laboratory Improvement Amendments (CLIA) level validation that most university research settings simply cannot match. Projects are not expected to fend for themselves; they draw on shared infrastructure, shared expertise, and a culture that treats translation as a first-class research objective. A formal institutional partnership with Brigham and Women's Hospital, part of the Mass General Brigham health system, ensures that technologies are, as our team's SWOT analysis noted, 'born' with a defined clinical unmet need. Researchers have a direct pipeline to patient samples, and feedback from clinical grand rounds shapes design decisions long before a prototype reaches a formal validation study. That's not how

[9] Wyss Institute for Biologically Inspired Engineering. (n.d.). *Wyss Institute*. Harvard University. https://wyss.harvard.edu/
[10] Ibid.
[11] Ibid.

most academic labs work, and the difference shows. Technologies emerging from Wyss tend to be, from day one, oriented toward real patients rather than theoretical ones. The institute also benefits from a hub-and-spoke organizational logic in which adjacent research programs can share findings, materials, and personnel without the siloed inefficiency typical of large research universities. This cross-pollination is not accidental; it is designed. Projects build on each other's existing work rather than stalling in the proverbial 'valley of death' that swallows so many promising early-stage technologies before they reach clinical or commercial relevance.

Interview

In the course of our team's background research, we had the opportunity to sit down with Dr. Michael Super, who leads the Advanced Technology team and plays a central coordinating role within the Wyss MedTech region. The conversation was more illuminating than any paper we had read up to that point, and a few themes emerged that shaped our subsequent project framing in important ways. On the question of how Wyss balances funding projects with obvious startup potential against those with less immediate commercial appeal, Dr. Super described a tiered funding architecture that allows for both near-term and longer-horizon investments. Different categories of support map onto different stages of technological maturity, and the institute's track record of more than 50 spinouts since 2009 suggests that this approach has real teeth. The diagnostics-versus-treatment question was one we had wondered about going in. Dr. Super pointed to the Diagnostics Accelerator (DxA), a dedicated Wyss collaboration led by David Walt and focused exclusively on diagnostics, as one pole of the spectrum. Theranostics, technologies that sit at the intersection of therapy and diagnostics, occupy an important middle ground. In practice, the balance depends heavily on faculty expertise and the specific technological opportunities at hand. What stood out most, though, was the candor with which Dr. Super described *operational friction*. When we asked where projects most often break down, he pointed, perhaps surprisingly, to the physical logistics of the institute itself. Moving materials between the Wyss's split campus locations (the Fenway-area Seaport Engineering Center and the Longwood Medical Area) creates real bottlenecks. Researchers and collaborators frequently don't know who is working on what across campus. The Lyft service that connects the SEC to the medical area, he noted, actually does not stop at the Wyss building. Dr. Super even mentioned that biological samples face strict transport protocols, with specific constraints on carrier vehicles and container requirements, creating friction that compounds across an already complex research operation. He also flagged stockroom and inventory management as an area of persistent, low-level dysfunction. Labs experience high demand for single pieces of equipment, scheduling conflicts arise, and there is no centralized, standardized system for tracking consumables and materials across teams. These may sound like small inconveniences, but in a research environment where delays cascade and reagents have shelf lives, poor logistics translate directly into wasted time, money, and scientific momentum. Dr. Super's Advanced Technology team ends up acting as an impromptu coordination layer, connecting faculty members who discover mid-project that their work is compatible.

SWOT Analysis

Our team synthesized findings from primary and secondary sources into a structured strength, weaknesses, opportunities, and threats analysis of the Wyss' MedTech region.

In terms of the strengths, the Wyss Institute has about 40 engineers on staff who have worked in the FDA or industry before, meaning they can build and test prototypes in-house without relying on outside help. They also have a formal partnership with Brigham and Women's Hospital, giving them direct access to real patient samples for research. Since 2009, they've spun out over 50 companies, including ReadCoor (acquired for $350 million) and Emulate, Inc. Their core patents cover Organ-Chips, CRISPR, and synthetic biology platforms. The way the institute is structured also lets projects share resources and build on each other's work, which helps promising ideas survive the funding gaps that kill a lot of early-stage research.

In terms of the weaknesses, one persistent problem is growing: getting something from a working lab prototype to certified, large-scale manufacturing is consistently difficult. Most of their spinouts also need $50–100 million or more to get off the ground, which makes them vulnerable when borrowing is expensive. The institute leans heavily on a small number of star researchers, particularly Ingber, Church, and Mooney, which creates real risk if any of them leave. On top of that, most of the institute's funding comes from outside grants rather than reliable internal revenue. Animal testing and regulatory approval timelines also regularly slow projects down.

In terms of the opportunities, DARPA and military funding give Wyss the ability to develop things like exosuits and wearables for defense first, then adapt them for everyday use later. The global population is aging, which is driving growing demand for home health monitoring tools and assistive soft robotics. Their expertise in biodegradable materials like silk and chitosan puts them in a strong position to lead in eco-friendly medical devices. Their Organ-Chip technology, when loaded with a specific patient's cells, could eventually replace traditional clinical trials for personalized drug testing. They also sit on unique, high-quality biological datasets that AI companies are willing to pay to license.

In terms of the threats, Federal budget cuts or a recession could quickly wipe out their NIH and DARPA grant funding, which they rely on heavily. Tech giants like Google Health, Apple, and Verily are constantly recruiting away top researchers, which is a steady drain on talent. Even when Wyss technologies are proven to work, hospitals and health systems tend to adopt new tools slowly, which delays commercial success. Regulatory requirements from the FDA and EU are also getting more complex and expensive, and spinouts can run out of funding before they ever get approved. And as their spinouts grow and become more valuable, they become bigger targets for patent lawsuits. Lastly, we learned that many of their devices depend on highly specialized chemicals and microchips that only a handful of suppliers make, and if those supply chains get disrupted by shortages, geopolitical tensions, or a single supplier going under, it could stall multiple projects at once.

A few points bear elaborating. On the strengths side, the foundry model and hospital partnership are genuinely unusual in academic research. Most universities are good at discovery but structurally indifferent to deployment; Wyss has built deployment into its operating logic. The IP portfolio, anchored by foundational platform technologies rather than single-use devices, enables multiple licensing revenue streams from individual core discoveries, which is a more durable commercial model than it might first appear.

The weaknesses, however, are real and worth taking seriously. The scale-up problem is, if anything, underappreciated. Moving from a lab prototype to ISO 13485-compliant mass production is a fundamentally different engineering challenge than the original research, and the institute's current approach of engaging external manufacturing partners on a project-by-project basis lacks standardization. This is a structural gap, not a personnel problem.

The opportunities section is exciting. The convergence of aging demographics, biodegradable materials expertise, and AI-hungry tech companies looking for clean biological data creates an unusual set of strategic options for an institute with Wyss's capabilities.

The threat landscape is more sobering: regulatory complexity is rising, talent poaching is relentless, and federal funding has become less predictable. Supply chain risks for specialized reagents, something that barely registered as a concern before 2020, now deserve a real spot in risk planning.

1.3.4 - Women's Health Division

Introduction to Women's Health

The Women's Health workstream examined how the Wyss Institute supports the translation of women's health research from early scientific discovery toward clinical, commercial, and implementation pathways. This area was selected because women's health remains historically underfunded and underdeveloped, despite significant unmet clinical needs across reproductive health, maternal health, gynecologic disease, and broader sex-specific health conditions.

At the Wyss, women's health innovation is supported through the Women's Health Catalyst, which helps position women's health as a strategic area for translational research.[12] The Catalyst benefits from the broader Wyss infrastructure, including internal funding mechanism, intellectual property support, legal guidance, business development resources, and access to interdisciplinary scientific expertise. These resources allow early-stage projects to generate preliminary data, clarify potential product pathways, and move closer to real-world application.

However, women's health translation also presents unique challenges. Projects often face pressure at the point where scientific discovery must become a clearly defined product. At this stage, teams must balance clinical need, market opportunity, fundability, and stakeholder interest. Women's health indications can be especially vulnerable to being redefined or deprioritized if they are perceived as less attractive to traditional venture capital, pharmaceutical, or commercial partners. As a result, this workstream focused not only on the strengths of the Women's Health

[12] Wyss Institute for Biologically Inspired Engineering. (n.d.). *Women's health*. Harvard University. https://wyss.harvard.edu/technologies/womens-health/

Catalyst, but also on the structural barriers that can limit translation, collaboration, and long-term impact.

Key Projects and Focus Areas

The Women's Health Catalyst supports a range of projects that address historically underdeveloped areas of women's health. These projects demonstrate the breadth of the Catalyst's work, spanning breastfeeding and lactation, menstrual health, fertility preservation, and ovarian cancer therapy development.

One notable project is Lactation Biologics, which focuses on increasing milk production to support breastfeeding and improve infant health outcomes.[13] This project reflects the Catalyst's emphasis on areas of women's health that are clinically important but have traditionally received limited commercial and research attention.

Another key project is the Heavy Menstrual Bleeding Organ-Chip, which uses organ chip technology to model heavy menstrual bleeding and improve understanding of a condition that affects many women but remains under-studied. This project is especially aligned with the Wyss' broader strengths in biologically inspired engineering and translational platform technologies.[14]

OvaVasc is another important women's health project focused on fertility preservation. By addressing ovarian tissue engraftment and reproductive health, OvaVasc reflects the

[13] Wyss Institute for Biologically Inspired Engineering. (n.d.). *Lactation biologics: Increasing milk production for healthier babies*. Harvard University. https://wyss.harvard.edu/technology/lactation-biologics-increasing-milk-production-for-healthier-babies

[14] Wyss Institute for Biologically Inspired Engineering. (n.d.). *Wyss Institute receives Wellcome Leap funding to develop first human Organ Chip model for heavy menstrual bleeding*. Harvard University. https://wyss.harvard.edu/news/wyss-institute-receives-wellcome-leap-funding-to-develop-first-human-organ-chip-model-for-heavy-menstrual-bleeding/

Catalyst's interest in supporting technologies that can improve quality of life and expand treatment options for patients.[15]

The Women's Health Catalyst also connects to emerging work in ovarian cancer therapy development. Interview insights highlighted the relevance of immune-system in vitro models and organ-on-chip approaches in evaluating immunotherapies and vaccines, particularly in ways that can inform pharmaceutical decision making. Together, these projects show that the catalyst is not limited to one narrow clinical area. Instead, it supports a broader translational ecosystem aimed at advancing women's health innovation across multiple stages of research and development.

SWOT Analysis

Research for the Women's Health workstream began with a SWOT analysis to establish the strategic position of the Women's Health Catalyst and identify key external landscape drivers shaping women's health translation. The analysis examined funding dynamics, market incentives, adoption pathways, translational infrastructure, and collaboration barriers. This initial synthesis helped the team identify the highest uncertainty assumptions, including where operational bottlenecks occur, what partnership types most accelerate translation, and how women's health projects are prioritized as they move toward product definition.

The SWOT analysis identified several core strengths. The Women's Health Catalyst benefits from the Wyss' internal funding, IP and legal infrastructure, business development support, and interdisciplinary research environment. These resources can accelerate early stage

[15] Wyss Institute for Biologically Inspired Engineering. (n.d.). *OvaVasc: Enhanced ovarian tissue engraftment for fertility*. Harvard University. https://wyss.harvard.edu/technology/ovavasc-enhanced-ovarian-tissue-engraftment-for-fertility/

work by helping teams generate preliminary data, clarify commercializations pathways, and maintain a line of sight toward clinical or commercial application.

At the same time, the analysis surfaced several weaknesses.Women's health projects can be affected by siloed lab structures, unclear shared incentives, and limited data sharing across teams. These issues can make collaboration dependent on individual goodwill rather than formal systems. Without clear credit allocation or shared incentives, researchers may have limited motivation to coordinate across labs, which can lead to duplicated effort and slower translation.

The SWOT analysis also highlighted major opportunities. Women's health is receiving increased attention from funders, policymakers, researchers, and commercial stakeholders. This creates an opening for the Catalyst to support projects that may have been historically overlooked. The Wyss' translational model is especially well positioned to help these projects move from early discovery toward more defined clinical and commercial pathways.

However, the analysis also identified important threats. Women's health projects often face funding and market pressures that can distort their original clinical purpose. In some cases, a project may be pushed toward a different indication that appears more fundable, even if that shift moves it away from its original women's health target. Projects focused on underserved populations may also struggle to attract traditional venture capital or pharmaceutical investment, even when the technical case is strong.

Interview

Following the SWOT analysis, interview questions were developed to test and refine the team's findings through direct insight. The team prepared and conducted an interview with Dr.

Girija Goyal, an immunologist developing immune system in vitro models and assays, including organ-on-chip approaches, to evaluate immunotherapies and vaccines. Her work informs pharmaceutical decision making, and she recently received an ARPA-H Women's Health Sprint Spark award connected to ovarian cancer therapy development.

The interview reinforced the importance of operational accelerators within the Wyss model. Internal funding mechanisms can help early-stage women's health teams generate preliminary data, while business development and IP/legal support help keep downstream application and commercialization considerations in view. This infrastructure gives women's health projects a stronger translational foundation than they might have in a traditional academic lab setting.

The interview also emphasized that women's health capacity is enabled by both funding and talent formation. Some efforts emerge organically from researchers who were originally recruited for other areas, but who develop women's health side projects that later mature into fundable programs. This suggests that the Catalyst's long-term value may depend not only on formal project funding, but also on creating an environment where researchers can explore women's health applications within broader scientific programs.

A major theme from the interview was the complexity of cross-team collaboration. Collaboration is strongest when incentives align and when projects maintain clear stakeholder pull. Some of the most successful women's health examples involve close linkage between patients, physicians, and researchers. However, translation can stall when teams lack aligned incentives, clear credit allocation, or long-term commitment to shared goals.

The clearest bottleneck identified in the interview was the transition from fundamental science to product definition. At this stage, projects must move beyond scientific proof of concept and define a clinical target, user need, product pathway, and funding strategy. This is where market-positioning pressures often appear. In women's health, there can be pressure to re-scope a product toward a different indication that appears more fundable, even if the original indication more directly addresses an unmet women's health need.

The interview also surfaced prioritization constraints and data sharing gaps. With many concurrent ideas, prioritization tends to weigh strength of evidence alongside perceived market opportunity. However, the interview also noted the importance of reserving space for high-impact projects that may not have obvious financial partners. Structural barriers to collaboration, including duplicated work, siloed datasets, and credit-allocation dynamics, can make it harder for teams to share data and coordinate efficiently.

Finally, the interview highlighted a recurring tension between impact and returns. Women's health efforts aimed at underserved populations can be difficult to finance through traditional venture capital or pharmaceutical models. In these cases, mission-aligned nonprofit, philanthropic, or government support may be especially important. This creates a practical constraint on translation pathways, even when the technical feasibility of a project is strong.

Together, the SWOT analysis and interview insights helped validate which constraints are primarily operational and which are driven by external market conditions. Operational constraints include collaboration barriers, data sharing gaps, unclear incentives, and prioritization challenges. External constraints include fundability pressures, limited traditional investment pathways, and the historic underfunding of women's health. These findings grounded the team's recommendations in specific Wyss mechanisms, including funding and IP support, collaboration structures, and partnership strategy, rather than abstract sector-level observations.

1.3.5 - Diagnostics Division

Introduction

The Diagnostics department at the Wyss Institute operates with a mission focused both on access to diagnostic tools for low-income populations and innovation in its field. Instead of developing isolated diagnostic tools, the Wyss' diagnostics branch focuses on creating broadly applicable platforms with multiple diagnostic tools to address multiple clinical and environmental problems. This approach starkly differentiates itself from a typical diagnostics innovation pipeline as the institute focuses on a clinical indication and tries to match it with a technology that can address that exact unmet need. This results in a patient-first philosophy that is embedded within this department and results in a uniquely positioned design at the intersection of deep academic research, real-world clinical validation, and commercial translation through the Wyss Diagnostics Accelerator.

Key Projects and Focus Areas

United by its overarching commitment to making powerful diagnostic tools accessible across clinical and non-clinical settings, the diagnostics branch has a large portfolio that spans human health, women's health, environmental monitoring, and global equity.

Within its most impactful work, we have the Point-of-care and Rapid diagnostics technologies such as NeoSense, COPDx program, and the Blood Clot Dx platform. These tests focus on using small samples of saliva and other readily available biomarkers to detect sepsis, COPD, and even patients at risk for blood clots before they happen. Expansion of these efforts into low-resource and instrument-free diagnostics tools that can be deployed in remote or under-resourced environments without specialized equipment reflect the branch's commitment to health equity. A champion of this focus area has been the Instrument-Free Molecular Diagnostics technology, a synthetic biology-based molecular diagnostics platform that enables creation of low-cost, highly accurate tests for non-clinical settings through freeze-dried, cell-free reactions.

On the other hand, Environmental and Planetary Health Diagnostics mark an exciting and expanding frontier for the branch, championed by technologies already mentioned in the Sustainability overview like PFASense. This technology exemplifies the Wyss' unique role as a catalyst of cross pollination between departments and fields of research, like Diagnostics and Sustainability.

SWOT Analysis

Our SWOT analysis of the Wyss Diagnostics branch, informed by the Institute's public materials and later our interview, surfaced a set of structural strengths, weaknesses, opportunities, and threats that are noteworthy and can help guide our problem selection.

On the strengths side, this branch uniquely benefits from the dedicated and well-organized patent structure, along with a full-time Wyss patent attorney, an Advanced Technologies Team (ATT) that pairs industry experience alongside research talent, and an "Entrepreneurs in Residence" program that all work together to push discoveries toward commercialization/spinning off rather than stalling in academia. The Wyss track record with high profile clients like DARPA validates the quality and advancements of the science produced in the institute, while also attracting philanthropic funding. Diagnostic's core mission of targeting underserved populations and under-resourced diagnostic issues also attracts governmental investment, creating meaningful funding diversification.

The weaknesses, however, reflect a symptom of this unique translational focus which is the lack of a programmatic system for synthesizing ideas across the Wyss to bolster ideation processes and increase the odds of project success inside the institute and outside. This lack of internal coordination can further exacerbate the impending threat of federal funding instability and its unconformable reliance on governmental grants for large multi-year projects. This instability can make external funding sources and incubators increasingly attractive to Harvard-affiliated researchers, creating a talent retention challenge.

The opportunities are significant and well-timed, as the branch is naturally positioned for rapid-response work for pandemics, environmental toxins, and emerging infectious diseases. The growing interest in AI-based tools for combining and generating new project ideas could help address the current over-reliance on individual people for intellectual synthesis. Furthermore, AI/ML applications in the biotech realm have increased rapidly for therapeutic development, while diagnostics still lags which could place the Wyss as a pioneer, especially given its already on-going projects in this field.

Interview

An interview was conducted with Rushdy Ahmad, PhD, Director of the Wyss Diagnostics accelerator. This interview yielded key takeaways about the department's largest challenges. As expected, this field has highly technical challenges that are beyond ES96's focus, like optimizing biomarkers analysis in their devices or optimizing their delivery pipeline for their diagnostics platforms to make them even cheaper for low-income populations. However, addressing the Wyss' ineffectiveness in actively promoting project cross pollination on the researcher scale (and not necessarily Principal Investigator driven) through some system could be beneficial for the community.

1.3.6 - Operations Division

The operations division of the Wyss Institute encompasses all of the different systems that enable the Wyss to complete its work. To better understand the operations department of the Wyss, the team worked to identify different strengths, weaknesses, opportunities, and threats that the division is currently facing to determine how to create solutions that are of benefit.

In terms of the strengths, the Wyss excels in its access to highly qualified faculty, which provides a unique translational model focused on comprehensive support infrastructure. In addition, the Wyss provides a substantial grant-focused team that is dedicated to helping the research have the necessary support for success. For the weaknesses, due to the nature of the research, many of the Wyss' operations are dependent on continued governmental and philanthropic funding from different grants. In addition, due to the high amounts of technicality that is required by the projects, many researchers feel a strong pull to industry, meaning that it is difficult to continue the longevity of the team. For the opportunities, there are places to expand on the operations of specific departments, such as the Women's Health, Sustainability, and climate tech divisions, due to the current expansion of these particular markets. Lastly, in terms of some threats that the operations division could be facing, these include regulatory uncertainty for specific areas of research, funding volatility for biotech, talent retention challenges, and intellectual property disputes.

In addition to the different deep dives of the Wyss, the ES96 team completed an in person visit to the Wyss campus to gain a better understanding of the day-to-day operations of the researchers. The Wyss Institute consists of three floors that contain various laboratories, offices, and high-specialty equipment that enable the researchers to complete their groundbreaking work. When the team was visiting the Wyss, we were able to speak to multiple researchers who presently encounter daily challenges at the institute. In particular, two challenges were brought up - mainly regarding the transportation of biomaterials between labs, and the current billing practices regarding the stockroom of supplies for general usage across all researchers.

For the transportation issue, researchers are physically located across the different campuses of Harvard and the Wyss Institute as well. Safe transportation of biomaterials that are utilized in many of the research projects across the different campuses is a complex issue due to a lot of the legal legislation. And for the stockroom, the operations team manages all of the shared resources, including the stockroom, which has general-purpose supplies used across all research divisions, providing general lab materials. The current billing model charges per visit and is averaged across all labs, causing difficulties as groups are not accurately charged for what they take.

To better understand the challenges faced by the Wyss' operations division, we were able to have a meeting with Mike Carr, who is the director of operations at the Wyss. In our conversation, we were able to better understand the current state of the stockroom system and develop the different criteria necessary to better design the system.

Thus, the deep dive and research into the operations division highlighted challenges that are encountered on a daily basis by researchers, and gave the ES96 team a better understanding of the problem spaces that could be tackled with our work

1.4 - Problem Statement

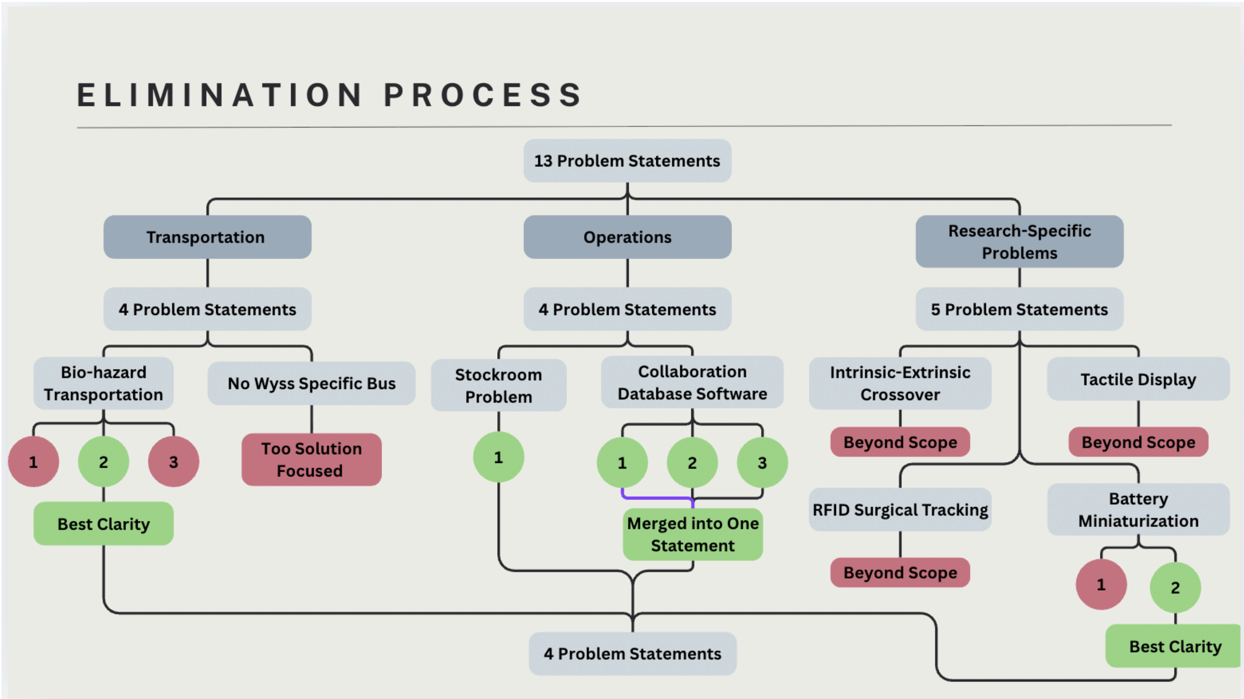
1.4.1 Ideation and Synthesis

Following the research and client interviews across the different divisions of the Wyss Institute, we discovered several problems that affected all of the researchers at the Wyss Institute. There were, additionally, some identified problems that affected specific researchers and departments at the Wyss Institute instead of the institution as a whole. Broadly, these problems were sorted into three categories: transportation, operations, and research-specific. The transportation issues revolved around the movement of both researchers and biological hazards between different locations including the Wyss Institute, the Longwood Campus, and the Harvard University Campus in Cambridge, Massachusetts. The operations problems were those that affected the communication and coordination between Wyss labs and researchers and the issues with shared spaces and resources for Wyss laboratories. Lastly, the research-specific problems were those that were highly lab specific as opposed to problems that affected the client as a whole, and included T-Cell research with the Therapeutics division and wearable technologies with the MedTech division.

The transportation problem evaluated by the team were the difficulties with the transportation of biological materials. The Wyss Institute works in collaboration with a multitude of labs and locations across the Cambridge area. This results in the necessary movement of these biomaterials. Due to legal stipulations and regulations, as well as liability logistics, it is often the case that a researcher is required to drive a personal vehicle or walk with their samples and bio-materials in order to transport them between locations, which is both cumbersome and time consuming. The movement of researchers between different locations was determined to be a difficulty because of the lack of a dedicated form of public transportation for Wyss researchers, as the shared M2 Shuttle route along Massachusetts Avenue prioritizes general university riders and the poorly located stops and unpredictable travel times can lead to disruptions in work flow as well as delays.

The operations problems that were evaluated by the team were concerning an insufficient communication system between laboratories, the inefficiency for charges from the current Wyss wide storage room system for general laboratory materials, and the funding difficulties that arise for the Wyss Institute because of its reliance on the same funding pipelines as Harvard University. For communication, this is a problem that has a wide variety of presentations. The researchers at the Wyss Institute work fairly independently within their own lab, and this limits the cross-pollination between these labs that could lead to the synthesis of new projects or other collaboration efforts between these laboratories. Ultimately, this lack of potential communication could be harboring potential opportunities which could greatly benefit the Wyss. The grant issue was explored in conjunction with the federal withholding of grants that occurred earlier this year from the Trump administration. Since the Wyss works with the government for grant funding, this withholding manifested in budgetary constraint issues for the institute, leaving many labs and projects scrambling for funds. As for the last operational issue, the Wyss wide lab supply room was identified. This problem was identified on an investigation trip to the institute. Although lab specific supplies are ordered specifically by researchers. General supplies such as pipette, tape, paper, etc… are generally ordered in bulk for the use of the entire institute. This is currently managed on a per-visit basis payment plan where researcher swipes are used to adjudicate cost. In essence, if someone swipes in and grabs a box of pencils, and another swipes in and grabs a multitude of materials, the same would be charged to their respective labs. This unequal charging is what we determined to be a problem within the Wyss as this is not an efficient way to manage lab supplies.

1.4.2 Elimination Process



To begin the problem statement synthesis process following problem identification, the team split into three groups each tasked with developing several problem statements for the identified problems. Some teams developed the different problem statements for a singular issue, which proved to be critical for viewing each problem at a wider scope and bringing together multiple perspectives to evaluate the core of each difficulty faced by the Wyss. After the development of thirteen problem statements, the team had a collaborative discussion in order to work through the elimination process down to three primary problem statements, which were then presented to the client.

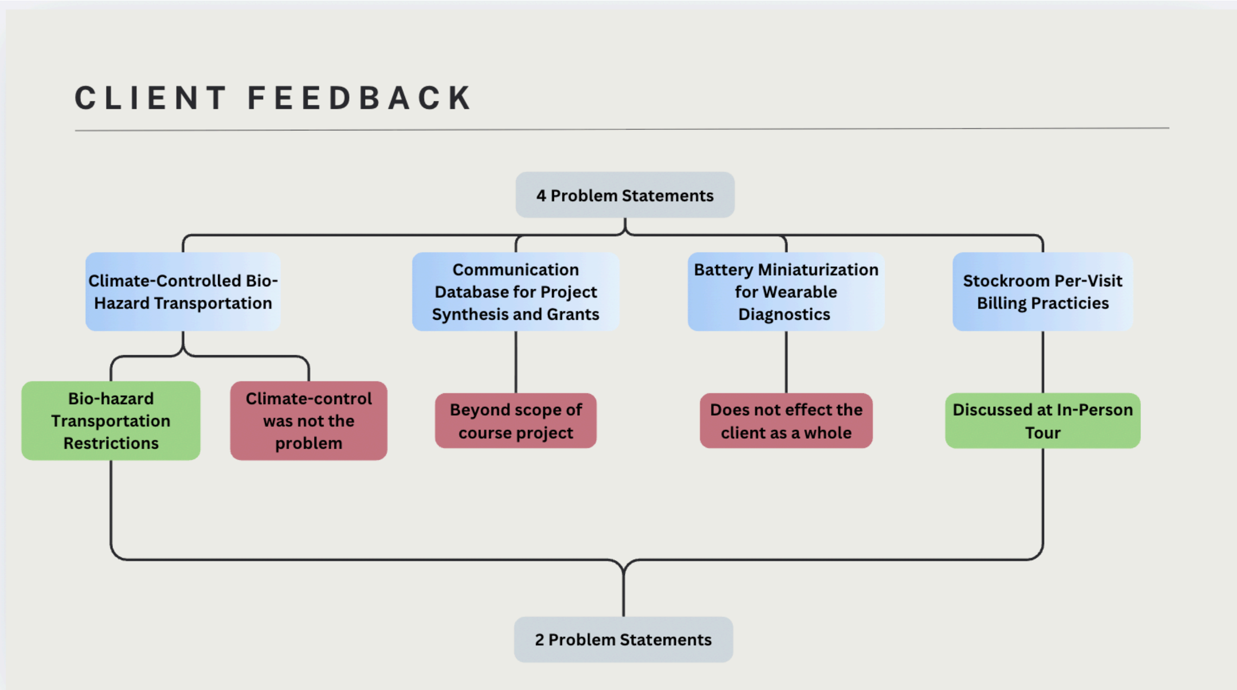
For the aforementioned problems in transportation logistics, operations logistics, and lab-specific research, there were a wide variety of reasons that certain problems were eliminated. Of the thirteen written problem statements, four of them were for the transportation logistics issues, with three of the written problem statements being for the same problem of hazardous biomaterial transportation. Of these three problem statements, one was chosen to be the primary that would be later refined for client presentation due to it having the best clarity and depth. The fourth transportation problem statement was eliminated because it was decided that the transportation logistical issue of the movement of researchers could potentially be folded into the other problem statement, and in elimination a note on the emphasis of time delays and work flow disruption was appended to the overall transportation statement.

The next three problem statements that were evaluated were operation logistics problems, and again they all focused on the same core issue – a lack of centralized communication between the different researchers and departments. With these statements, as opposed to choosing one that best encapsulates the problem, aspects of all three were merged together in order to create a new problem statement that best framed the dilemma with the most clarity. In the assimilation of these, a discussion on the root issue for this problem statement arose, as the group decided whether this was a predicament that could be fully addressed with a singular problem statement. The two potential problems that were discussed were the lack of an efficient coordination system and the limit on project synthesis inspired by pre-existing research, and more so whether these were separate problem statements or if a singular problem statement could concisely consolidate the issue as a whole. It was decided that the lack of project synthesis was a byproduct of the lack of communication between researchers and project teams, and moreover, it was decided how on a wider scale the absence of an efficient method of communication between laboratories also covered problems regarding grant applications, project collaborations, project research, and could be expanded to a global scale in regard to collaboration between different research departments and institutions across the globe.

Additionally, operations gave rise to another issue which came much later on during our synthesis process. On the investigative trip to the Wyss, the issue of per-visit billing was identified. Due to the tardy nature of this discovery, the group came up with this problem statement as a team. The statement was honed and then presented to the client.

The last area was research- specific problems. The team tasked with these problem statements came up with 5 different ones with potential. From these arose some strong contenders, most notably, the extrinsic-intrinsic cross over problem in therapeutics, and battery limitation issues in the medical technology group. However, ultimately after group discussion, most were eliminated in their entirety due to scope with only the battery limitation in size and duration being the only problem statement worth pursuing. This problem statement was then refined for later client presentation.

1.4.3 - Client Presentation and Project Selection



The four remaining problem statements were presented to the Wyss Institute during a client feedback session. Climate-controlled bio-hazard transportation, a communication database for project synthesis and grants, battery miniaturization for wearable diagnostics, and the stockroom per-visit billing problem were each evaluated with the client.

Two of the problem statements were eliminated quickly. Battery miniaturization was ruled out on the grounds that it did not affect the client as a whole, applying only to a narrow subset of research teams. The communication database was determined to fall outside the scope of the course project and did not seem suitable. Bio-hazard transportation, while a genuine operational pain point, revealed itself upon further research to be a regulatory dead end. Compliance requirements from MASCO, the Department of Transportation, and Harvard's internal bio-hazardous material transport policies precluded any viable engineering solution that was worth pursuing. To affect change here it appeared more of a long term political project around the changing of current legislation.

The stockroom billing problem was the clear and unanimous final selection. It had surfaced directly during the team's in-person visit to the Wyss, affected every research division significantly, carried a measurable and demonstrable outcome, and was within the scope of engineering design for our team. Dr. Michael Super confirmed its relevance, and the team committed to this problem as the sole focus for the remainder of the course.

1.4.4 Problem Statement

To bring together our findings into one problem statement, the team considered how each element of the problem classifies as the core issue, its cause, the affected population, or the desired outcome. The problem arose from the stockroom's billing structure: labs across the Wyss are charged on a per-visit basis averaged across all users, meaning that labs drawing small quantities of materials are charged the same as high-volume users. This is compounded by a lack of item-level tracking and largely manual inventory management, which forces researchers to visit multiple stockrooms to locate materials that may not be uniformly stocked. The affected stakeholders are all research labs operating within the Wyss Institute. The desired outcome is a restructured checkout system in which labs are charged only for the materials they actually take and use, supported by accurate, real-time inventory tracking. Thus, our investigate phase culminated in the following problem statement:

Shared stockrooms at the Wyss Institute use per-visit billing practices and sub-optimal inventory management processes, which cause labs to experience financial inequity and operational downtime.

Affected Stakeholders: Wyss Institute research labs

Desired Outcome: Accurate, usage-based cost allocation and real-time inventory tracking across all stockrooms

Problem: Labs are billed disproportionate amounts relative to the materials they actually consume

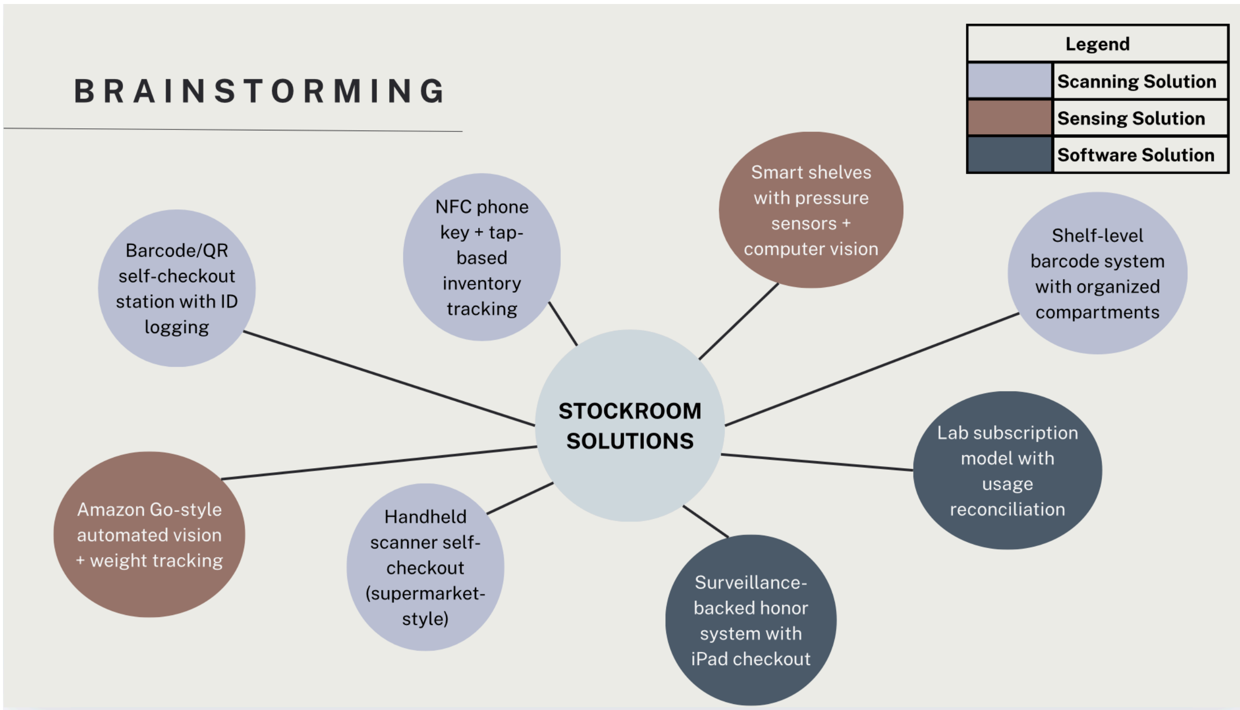
Cause of the Problem: The current billing model charges per visit, averaged across all labs, with no item-level tracking to differentiate usage

2 - Solution Ideation

Once the problem statement had been determined, the team began to design potential solutions that would effectively ensure that the issues that were described in the problem statement were sufficiently solved. After the initial brainstorming, several ideas were pulled together to consolidate the final solutions down to three main solutions, after identifying three clear categories that the solutions fell under. To determine the final solution that would move forward, a decision matrix was created based on specific criteria of success. This final solution was then presented to our client at the Wyss to receive feedback and receive suggestions about the design. All of these parts together enabled us to develop the final solution design, which was then built out in the prototyping stage.

2.1 Project Brainstorming

2.1.1 Initial Ideation Phase: Seven Ideas



To begin with the initial design ideation, the team came together to develop different implementation ideas to tackle the problem statement. Key ideas, as seen in the image above, were summarized into three different categories - scanning solutions (QR, barcode scanning, NFC), sensing solutions (smart shelves, automated vision/weight tracking) and software solutions (lab subscription model, app checkout). These solutions aimed to clearly and effectively design an intuitive system that could be easily utilized by researchers and administrators.

2.1.2 Idea Reduction to Three Ideas

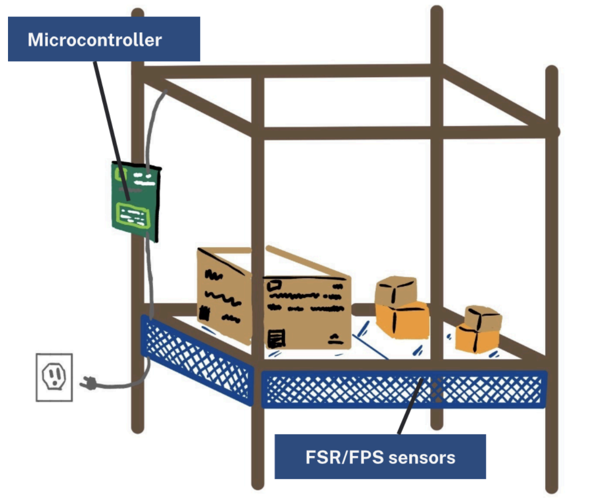
Based on the brainstorming session, the team evaluated all seven proposed solutions. Using criteria such as cost, ease of adoption, and robustness for error, the pool of viable projects was condensed down to three viable candidates.

The Amazon Go vending machine style was eliminated due to its cost and implementation complexity. Although an ideal way to disperse lab supplies to researchers, the requirements in computer vision, not to mention, the financial burden was immense and seemed unnecessary for the problem at hand. Ideally the problem is solved using the simplest idea possible to allow for the most seamless integration.

The handheld scanner and barcode system was consolidated into the NFC tag system. The team determined that the database and the subsequent back-side website was a great idea. It offered an intuitive and simple way to track transactions for finance, but when it came to correct usage, a strict software solution left the burden on the researcher. The team felt the researcher needed to be incentivized to use the system. Additionally we needed a design that could encompass the pre-existing security system with one that could integrate with the current Harvard University identification. Similar concerns were raised with the Ipad checkout as there was no integrated security protocol and the burden of proof still was on the researcher.

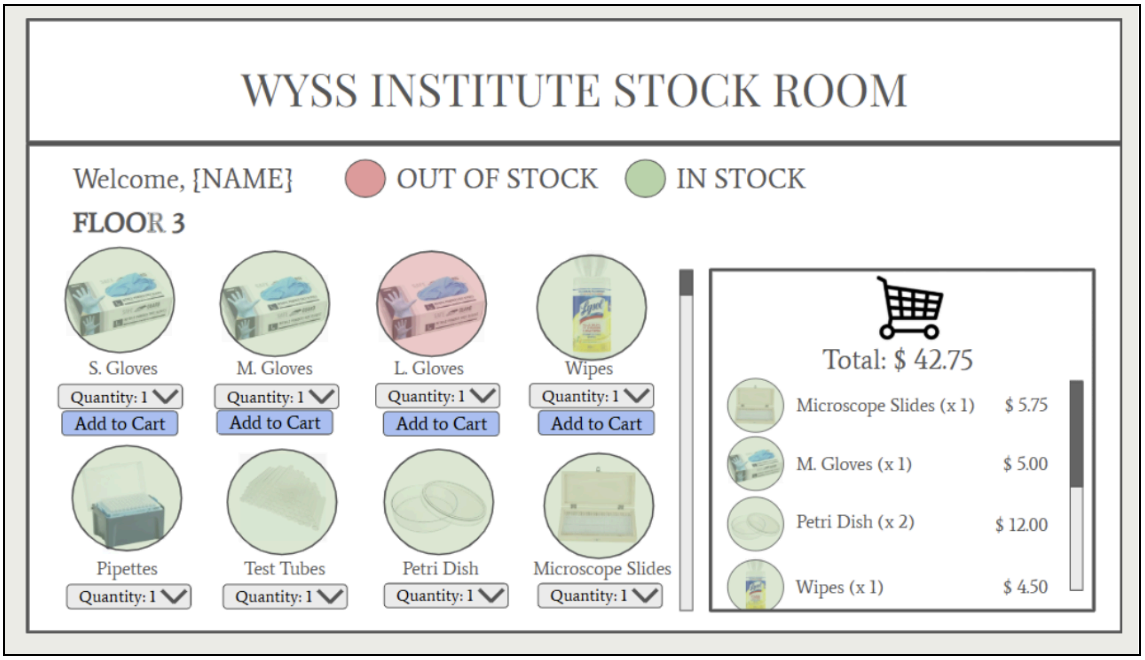
This left three strong candidates for a potential solution. The pressure plate solution with FSR/FPS sensors, a software only solution that would be an app interface, and a NFC based phone system with the hardware centralized to the stock room. These were the solutions that were subsequently evaluated for the decision matrix.

2.1.3 Solution One: Pressure Plates



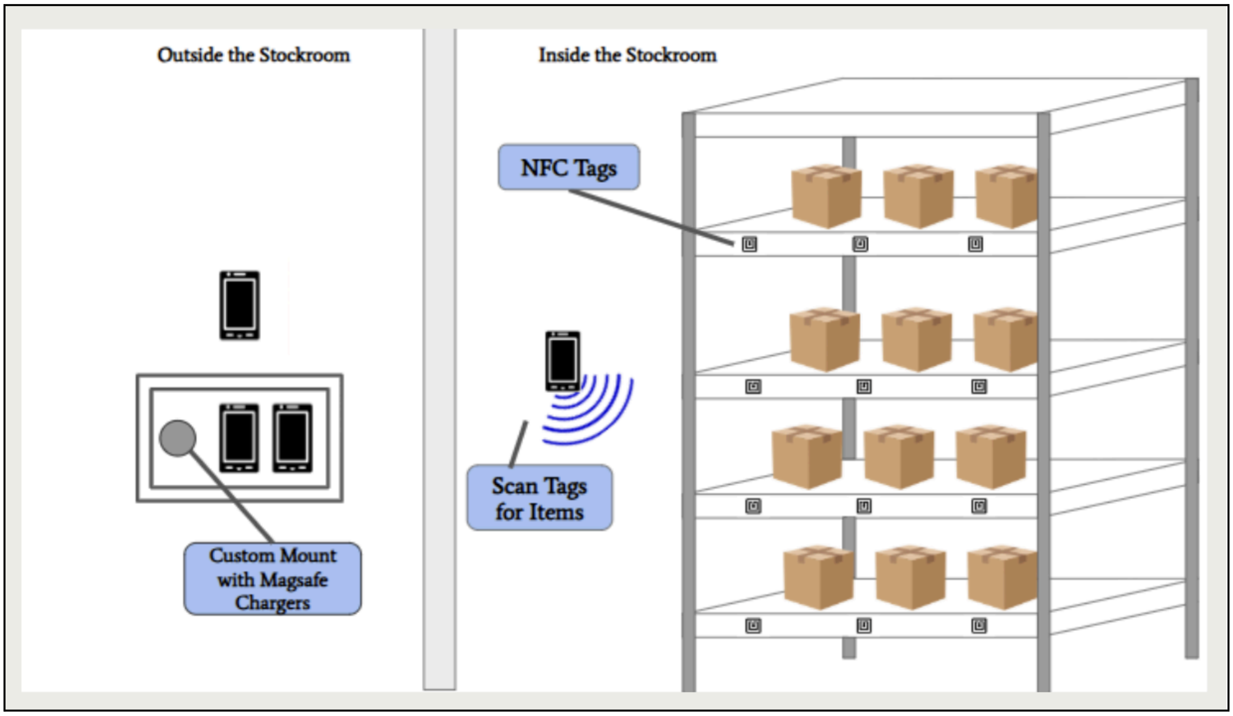
The first solution proposed was the pressure plate sensor inventory tracking system using a Force Sensitive Resistor (FSR) and Force Plate Sensor (FPS) technology embedded within the stockroom shelving. Each shelf would house a pressure sensing mat and have items allocated on individual mats connected to a central microcontroller. The microcontroller then detects changes in weight from the initial weight of the shelving, completing subsequent calculations to determine the total items removed based on the change. This system would allow for real time tracking, leaving minimal room for error when researchers retrieve supplies from the stockroom.

<u>2.1.4 Solution Two: Software Solution</u>



The second solution was a software-only approach that would allow researchers to log their item selection through a digital interface, an app that would be downloadable to the researcher. Upon entering the stockroom, the researcher would log in to their app on their phone, which would already have all admin information, such as HUID, labs, and grants that researcher is using. From there, the researcher would interact with a digital inventory that updates in real time as transactions occur. Once finished, the user would checkout using the cart interface in the app, the subsequent app flow would let them associate the cost with the corresponding grant.

2.1.5 Solution Three: NFC Phone



The final solution centered on a Near Field Communication technology which allows for contactless, simple, and short distance data exchange. In this system, each of the three stockrooms would be outfitted with NFC tags on each item, which would be passive labels for the shelves. Researchers would authenticate into a dedicated mobile application using phones that are linked to the stock room specifically for management use. By tapping with their HUIDs, it instantly logs them into the system, from there the user would maneuver throughout the room tapping the mobile device on a tag in order to log an item into the cart. After once all items have been scanned, an intuitive checkout system would allow the user to associate the corresponding costs to the pertinent grant for billing. After completing the transaction, the mobile device would get returned to the stock room for the next user to use.

2.2 Decision Matrix

Once we had fully developed each of our three possible solutions, the team decided to conduct a decision matrix to determine which one we would move forward with. The first step we took was to list all the criteria necessary to differentiate each solution from the others. We continued by setting the weights for each criterion, adhering to a minimum vote of 50% to ensure enough agreement within the team. Finally, we assigned a value of plus, neutral, or minus to each criterion for each solution to complete the decision matrix.

2.2.1 Decision Criteria

The decision criteria we decided to use were the following: Cost, Backend Ease, User Experience, Time Efficiency, Robustness for Error, Adaptability, Adoptability, Loss Prevention, and Scope. Our definition of cost included the cost of production, implementation, and maintenance of the system. Ease of backend usage meant how easy it would be for operations and administration to track items and restock the stock room properly. User experience was defined as how easy and convenient the system was to use for the researchers. Time efficiency simply tracked how long it would take to use each system with faster being better. Robustness to error was defined as how easy it would be to make mistakes with the system whether it be with the organization of the stockroom or the billing mechanism. Adaptability looked at how simple it would be to implement the system in a new environment such as a new Wyss building or a remodeled stock room. Adoptability was focused on whether the users would be able to work with the new system better than the old one. Loss prevention outlined how good the system would be at tracking stock to make sure nothing gets lost. And lastly, scope just showed whether the collective background of our team was experienced enough to deliver each system to the best of our ability.

2.2.2 Decision Criteria Weight Assignment

DECISION MATRIX

Criteria	Weight	Solution 1: Pressure Plate	Solution 2: Software	Solution 3: NFC Phone
Cost	1	-1	1	0
Backend Ease	2	-1	1	0
User Experience	3	1	-1	0
Time Efficiency	2	1	0	1
Robustness for Error	3	-1	0	0
Adaptability	2	-1	1	0
Adoptability	3	1	1	1
Loss Prevention	3	1	0	0
Scope	1	1	-1	1

LEGEND	
Bad	-1
Neutral	0
Good	1

SCORES	
Pressure Plate	4
Software	4
NFC Phone	**6**

For the assigning of decision criteria weight, we used a scale from 1 to 3 (1 being the least significant and 3 being the most significant). In the least important category we put cost and scope. The decisions came down to how relatively low the cost of all the systems were and the importance of having the best system regardless of how challenging it would be to develop. In the most important category, we chose to put user experience, robustness for error, adoptability, and loss prevention. All four of these categories had the greatest impact on the daily user and on operations management of the system. The rest of the criteria fell in between these two groups in terms of relative importance.

The pressure plate solution scored very well on loss prevention and time efficiency since the passive nature eliminated any sort of interaction with the system on behalf of the researchers. However, it performed extremely poorly on cost, error robustness, and adaptability. The Wyss stock rooms have little room and introducing a bunch of pressure mats, microcontrollers, and the necessary wiring could make an already tight space even more restrictive. Additionally, the immense up front cost investment outweighed the potential benefit of a new-billing tracking system. Lastly, the addition of other stock rooms or reconfiguration of stock room supplies would require immense reinstallation effort for the Wyss.

The software solution scored very favorably on cost and backend ease as there is no hardware necessary for this type of engineering solution. However, in terms of robustness and user experience it scored the lowest. This combined with the complete lack of a security system on the stock room and no physical reinforcement made the team feel that the system was vulnerable to forgotten checkouts and user error. This made this particular solution limited in its effectiveness as a billing tool.

The NFC solution scored highest overall in the decision matrix performing well across user experience, time efficiency, adoptability, and scope. The tap based login provides a simple and familiar authentication mechanism, while the item-level NFC scanning ensures every transaction is recorded. The system is very adoptable and has a physical use mechanism which the team thought was important and necessary for the honor system. Additionally, the NFC tags are very simple to program and are neutral in cost relative to the software solution.

2.2.3 Final Solution

As seen through the decision matrix, the NFC phone solution emerged as a clear winner with 6 pts, with both the pressure plate and software solution scoring 4 points overall. The result made it clear that the NFC chip was the most intuitive and most robust solution that we could field in terms of user experience and ease of implementation.

The pressure plate solution, despite scoring well on user experience and time efficiency, was ultimately heavily affected by its poor scores on cost, backend ease of use, and adaptability. The sensitivity of the pressure plates to miscalibration and significant initial up front cost made this the less practical use for the Wyss Institute. In regards to the software solution, it performed well on cost and backend ease but was penalized on user experience and scope due to the lack of accountability in its mechanism. Without physical enforcement, the system's sole reliance on voluntary self reporting made it unacceptable from the risk perspective. Additionally, if transactions were to be unaccounted for, stockroom counts and the fidelity of the billing system would be extremely inaccurate.

The NFC phone's ability to combine security authentication, in addition to a seamless implementation for use, made it the most complete and non-invasive answer possible while still solving the issues with the per-visit billing structure.

2.3 Client Feedback

After developing the final solution design, the NFC phone was presented to the client for initial feedback. The main concerns focused on the user interface on the phones and how it would allocate items to different labs and grants. The initial design did not take into account that an individual researcher at the Wyss may be a part of multiple different labs and that each lab may have various grants attributed to it. This feedback was easily incorporated into the phone website design as part of the checkout process.

3 - Prototyping

3.1 Prototyping Process Overview

After selecting the NFC-based item scanning system as the solution, we began prototyping versions of the solution. The build can be broken down into three distinct components, which interact with each other in various ways. The first component is the app (front-end), which researchers will interact with on the provided phones in order to log in and out, make item selections and edits, select which grants to bill to, and complete all other tasks related to purchasing items from the stockroom. The app team also handled the NFC sticker programming and Harvard ID-related programming since these items interface directly with the user. The second component is the website (back-end), which includes the handling of all user and stock data and will be only accessed by Wyss stockroom administrators. The third component is the main hardware component, consisting of a wall-mounted wireless charging station that will be mounted outside the stockroom and contain and charge the phones when not in use by stockroom users. Three separate teams worked on these three components, but communication across teams ensured that functional requirements aligned across the entire project build, both physical and software-based.

3.2 Front-End: App Development

The front-end component of the Wyss Institute Inventory Management System is a native Android mobile application built using the React Native framework and the Expo managed workflow. Development proceeded across four iterative phases spanning the construction of a member identity database, an administrative management panel, NFC-based authentication, and a multi-laboratory checkout flow. This section documents the architectural decisions, implementation details, and key challenges resolved throughout the development lifecycle.

3.2.1 Technology Stack and Architecture

The application was developed using React Native 0.81.5 via Expo SDK 54, with TypeScript 5.1 as the primary language. This combination was selected to enable rapid cross-platform iteration whilst targeting the Android platform specifically, owing to the NFC hardware requirements of the project. The complete technology stack is summarised in Table 3.2.1.

Layer	Technology / Version
Framework	React Native 0.81.5 via Expo SDK 54
Language	TypeScript 5.1
Navigation	React Navigation v6 (Native Stack)
Persistence	AsyncStorage (@react-native-async-storage 2.2.0)
NFC	react-native-nfc-manager

Layer	Technology / Version
Typography	Expo Google Fonts — Oswald (400, 600, 700)
Build System	Gradle 8.14 / Android SDK 36

Table 3.2.1 — Application Technology Stack

Navigation between screens is managed by React Navigation v6 using a Native Stack navigator, which delegates transitions to the native platform layer for optimal performance. Application state that must survive process restarts — principally the member identity database — is persisted via AsyncStorage, a key-value store bound to the device's local storage. This architecture enables fully offline operation with no dependency on external servers.

3.2.2 Phase 1 — Member Identity Database

The first development phase established the foundational data layer: a persistent mapping between NFC card Unique Identifiers (UIDs) and Harvard University Identification (HUID) numbers. Because the UID-to-HUID relationship held by Harvard's access control infrastructure is not publicly accessible to third-party applications, an internal mapping database was constructed and maintained within the application itself.

A static seed file at src/data/huidDatabase.ts defines the MemberRecord interface, which encapsulates a member's card UID, HUID, full name, and laboratory affiliations. Two lookup helper functions — lookupByCardUid() and lookupByHuid() — provide linear-time retrieval against the in-memory seed array and serve as a fallback on first launch.

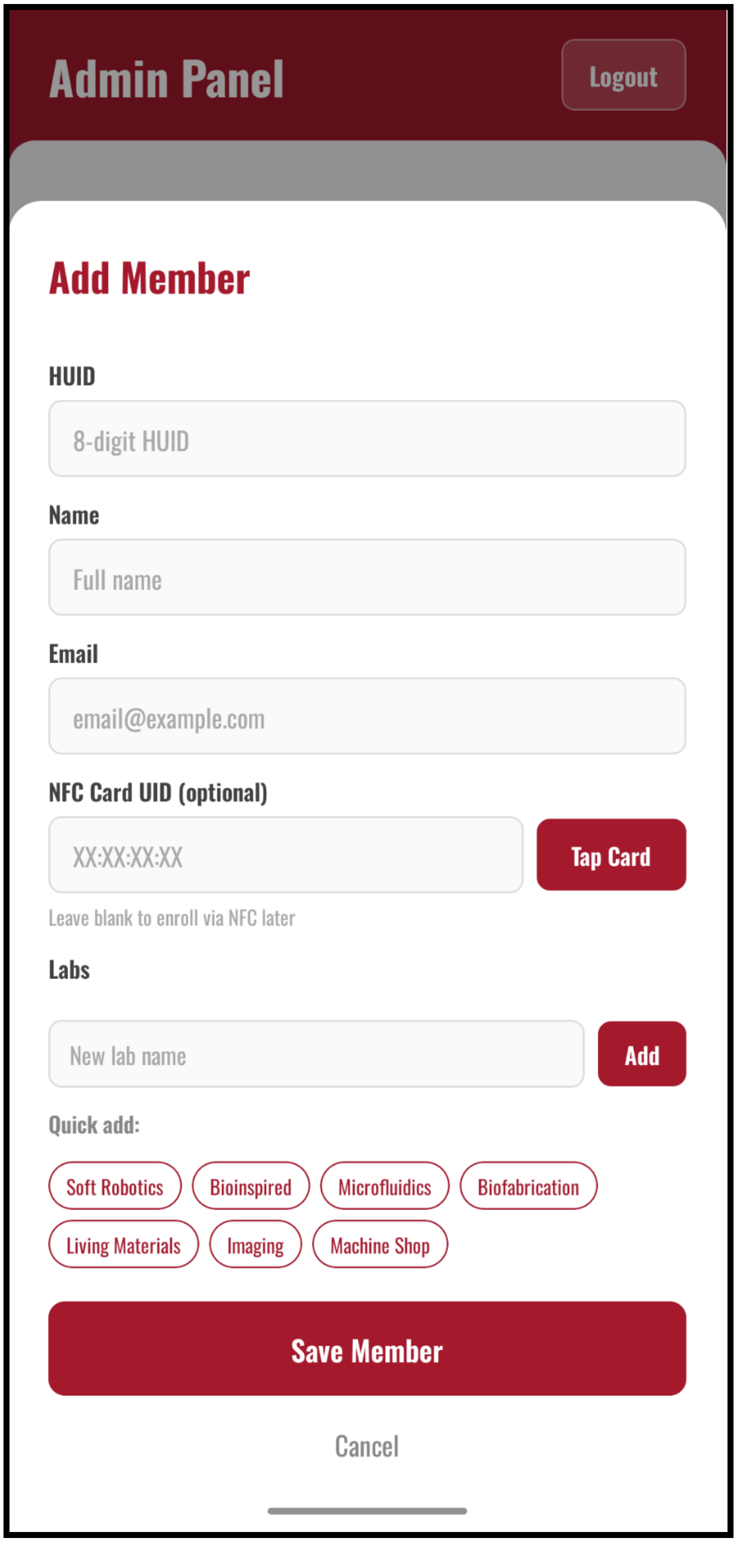
To allow runtime modification of the member roster without requiring application recompilation, a separate persistence layer was introduced at src/data/memberStore.ts. This module wraps AsyncStorage with typed CRUD operations: loadMembers() retrieves the stored member list, saveMembers() serialises it to JSON, and atomic helpers addMember(), updateMember(), and deleteMember() each load, mutate, and persist in a single call. This design ensures that all administrative changes survive application restarts entirely without a backend server.

3.2.3 Phase 2 — Administrative Panel

The second phase introduced a secure administrative interface for managing the member database at runtime. The panel is accessed by entering a PIN into the manual login field on the Login screen; successful validation routes the navigator to the Admin screen rather than the standard post-authentication flow.

The AdminScreen component (src/screens/AdminScreen.tsx) provides four principal capabilities. The member list view displays all enrolled members as card components, rendering each member's name, HUID, NFC card UID, and laboratory affiliations as coloured chip elements. Members who have not yet been enrolled with a physical card are flagged accordingly. The add member modal accepts a new member's details — including an optional card UID to be populated later — and validates against duplicate HUIDs before insertion. The edit member modal pre-populates all fields from the selected record and exposes a destructive delete action gated behind a confirmation dialogue. Finally, a laboratory management sub-interface offers seven preset laboratory designations as quick-add chips alongside a free-text field for custom entries.

The administrative panel adheres to the application's institutional visual identity: Harvard Crimson (#A51C30) is used as the primary accent colour, dark green (#4a5e1a) signals affirmative actions, and the Oswald typeface family is applied throughout. This design language was maintained consistently across all subsequent screens.



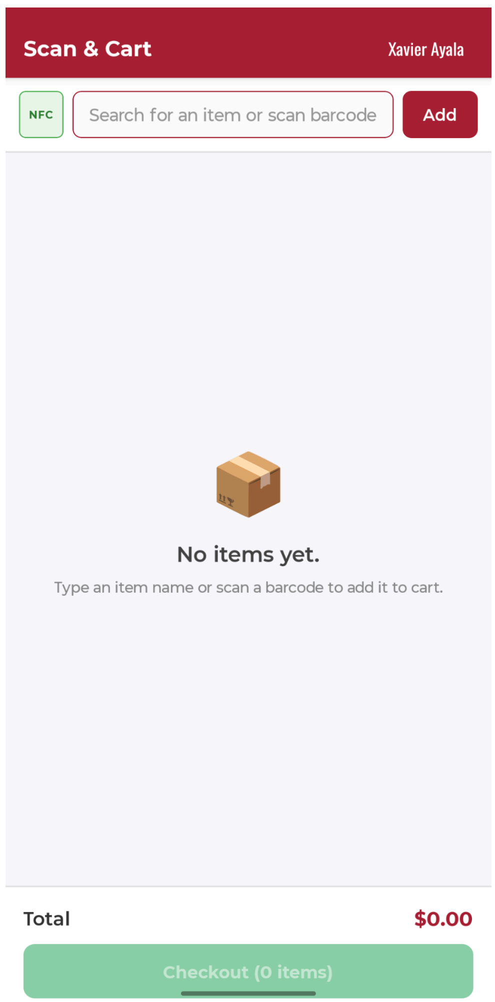
Prototype Admin Mobile Dashboard

3.2.4 Phase 3 — NFC Integration

The third phase integrated Near Field Communication support, which constitutes the core authentication mechanism of the system. The target credential is the NXP MIFARE Classic 4k card — the standard Harvard University Identification card — which operates at 13.56 MHz and presents to Android's NFC subsystem under two technology classes simultaneously: android.nfc.tech.NfcA (the ISO 14443-3A base layer) and android.nfc.tech.MifareClassic. The four-byte Unique Identifier broadcast by each card requires no cryptographic authentication and is therefore trivially readable by any NFC-capable Android device.

Because the react-native-nfc-manager library requires linkage against Android's NFC framework, it cannot operate within the Expo Go development sandbox; a fully compiled native build is required. A custom Expo config plugin was authored at plugins/withNfc.js to automate the two necessary modifications to the generated Android project during expo prebuild. First, the plugin injects a TECH_DISCOVERED intent filter into MainActivity within AndroidManifest.xml, allowing the Android NFC Tag Dispatch System to route incoming tag events to the application. Without this filter, the device presents an uninstructive system dialogue and does not forward the event. Second, the plugin writes res/xml/nfc_tech_filter.xml, which declares both NfcA and MifareClassic technology classes, instructing the dispatch system on which tag types the application is prepared to handle.

NFC lifecycle management is encapsulated in a reusable hook at src/hooks/useNfcScan.ts. An important implementation challenge was encountered during this phase: the initial approach of requesting the NfcA technology class via NfcManager.requestTechnology() proved unreliable for MIFARE Classic cards in practice — the device's NFC confirmation sound would play, confirming hardware detection, but the tag's identifier field would not be populated in the returned object. This was resolved by migrating to a passive event-listener model using NfcManager.registerTagEvent() and the NfcEvents.DiscoverTag event, which is agnostic to the technology class presented by the card. The UID is extracted from the tag's identifier field and formatted as a colon-delimited uppercase hexadecimal string for consistency with the database schema.



Prototype Design NFC Enabled Checkout Screen

3.2.5 Phase 4 — UI Redesign and Session Management

The fourth phase overhauled the application's visual identity and introduced session management behaviours appropriate for a shared kiosk deployment. Prior to this phase, the interface used a generic blue colour scheme with no institutional affiliation and no mechanism for automatically returning the device to an idle state after a completed transaction.

Typography and Visual Identity

The Oswald typeface was adopted as the application's primary typographic system. Its condensed proportions and strong geometric weight are well suited to a kiosk-style interface designed to be read at arm's length. Three weights were loaded at application root — Oswald_400Regular, Oswald_600SemiBold, and Oswald_700Bold — with the expo-splash-screen package configured to retain the splash screen until font loading completed, eliminating any flash of unstyled text on launch. All instances of the previous generic blue primary colour (#1a56db) were replaced with Harvard Crimson (#A51C30) across the Login, Cart, Checkout, and Navigator header components.

Login Screen and Post-Authentication Flow

The login interface was rebuilt to match a provided design specification. The redesigned screen features a full-screen Harvard Crimson background, the Wyss Institute wordmark loaded as a native image asset, a Harvard shield mark at the vertical centre of the screen, and a prominent NFC prompt rendered in Oswald_700Bold. A compact manual HUID input row is anchored to the bottom of the screen, comprising a translucent text field and a dark green submission button.

Prototype Design Template and Home Screen

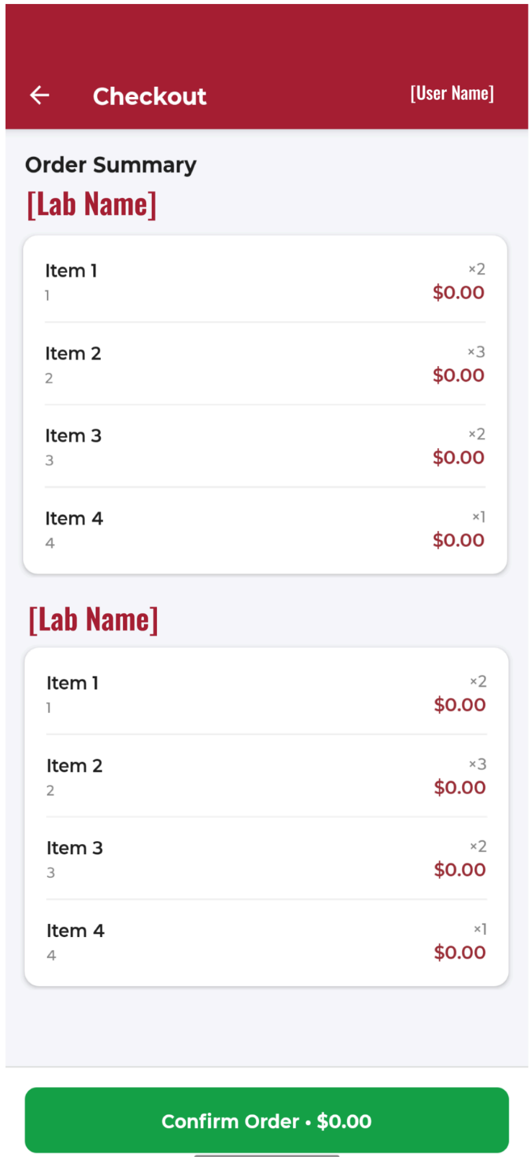
A dedicated LoginSuccessScreen was introduced to serve as a post-authentication transition. The screen presents a confirmation banner bearing the authenticated member's name, the Wyss Institute wordmark, and a large crimson circle carrying the instruction to tap the device on the door keypad. The circle entrance animation was implemented using React Native's built-in Animated API with useNativeDriver: true — a spring animation from scale 0 to scale 1 running in parallel with a 200ms opacity fade — ensuring the transform executes on the UI thread without competing with JavaScript execution.

Cart Interactions and Session Termination

Prototype of Wireless Access / Login Success Screen

The cart screen received a swipe-to-delete gesture implemented via the Swipeable component from react-native-gesture-handler. The interaction distinguishes two behaviours depending on drag distance: a partial swipe (less than 55% of screen width) opens the row to reveal a confirmable delete label, whilst a full swipe deletes the item immediately upon release. A critical implementation subtlety was identified and corrected during development: as the swipeable row snaps from the full-drag position back to the open position, the drag listener fires a final time with the settled value, which would reset the full-swipe flag prematurely. This was resolved by making the flag strictly monotonic within a single drag gesture — it can only be set to true during a drag and is reset only when the row returns fully to the closed position.

Two automatic session termination triggers were implemented to ensure the device reliably returns to an unauthenticated state following each transaction. An inactivity timeout navigates to the Login screen after fifteen minutes on the order confirmation screen. Additionally, connection to a power source — interpreted as returning the tablet to its charging dock — triggers immediate navigation to Login. Both subscriptions are torn down in their respective useEffect cleanup functions to prevent spurious navigation events.

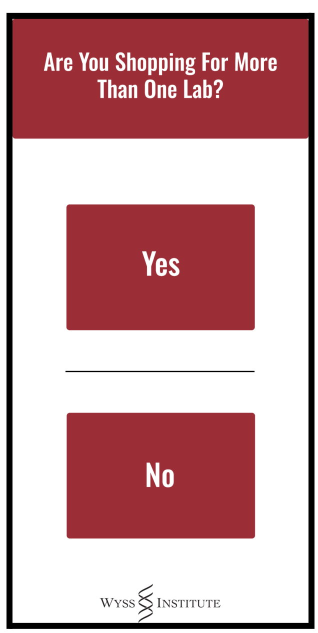


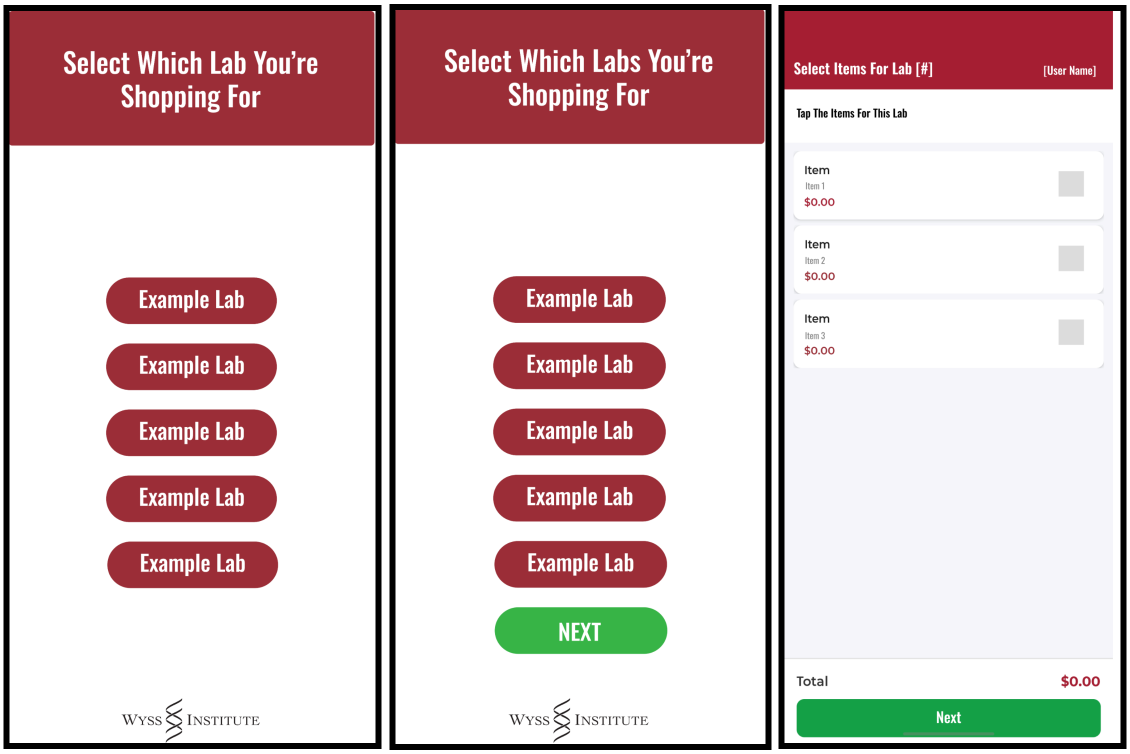
Prototype of Final Checkout Screen

3.2.6 Phase 5 — Multi-Laboratory Checkout Flow

The final development phase introduced a multi-laboratory checkout workflow to support members affiliated with more than one laboratory. Prior to this phase, tapping the checkout button sent all cart items to the CheckoutScreen as a flat list with no laboratory attribution. The new flow intercepts that navigation and guides the user through assigning items to one or more of their registered laboratories before confirming the order.

The navigation sequence is as follows. After reviewing the cart, the user is presented with a LabAssignmentQuestionScreen that asks whether they are shopping for more than one laboratory. Selecting 'No' routes them to LabSelectOneScreen, which presents a vertical list of pill-shaped buttons — one per registered laboratory — and immediately assigns all items to the selected laboratory before proceeding to checkout. Selecting 'Yes' routes them to LabSelectManyScreen, where toggleable pill buttons allow selection of two or more laboratories, with the continue button disabled until at least two have been chosen.





Prototype Design of Lab Separation Scheme

For multi-laboratory transactions, the user is cycled through a LabItemAssignScreen for each selected laboratory in sequence. Each iteration displays the remaining unassigned items as selectable cards; selected items are accumulated into the laboratory's assignment and removed from the pool before the next laboratory's screen is presented. Any items remaining after all laboratories have been processed are automatically appended to the final laboratory's assignment, ensuring no item is lost. The CheckoutScreen was updated to render the order summary in two modes: when laboratory assignments are present, items are grouped under bold crimson laboratory name headers; when assignments are absent, the original flat list is rendered, preserving backwards compatibility with direct navigation.

Several layout defects were identified and corrected during this phase. In LabSelectManyScreen, the use of alignItems: 'center' on the ScrollView content container caused child elements declared with width: '100%' to resolve against the shrunk content width rather than the screen width, resulting in a misaligned Next button and off-centre labels. This was corrected by changing the alignment to 'stretch' and repositioning the selection checkmark using absolute positioning so that button labels remained true-centre. A separate issue in LabItemAssignScreen, in which the laboratory name header rendered behind the device's status bar, was resolved by applying the top safe area inset dynamically using useSafeAreaInsets.

3.2.7 Build Environment and Technical Challenges

The transition from a managed Expo Go environment to a fully compiled native Android application introduced several build environment challenges, each of which required targeted resolution. Table 3.2.2 summarises the principal issues encountered and the remediation applied in each case.

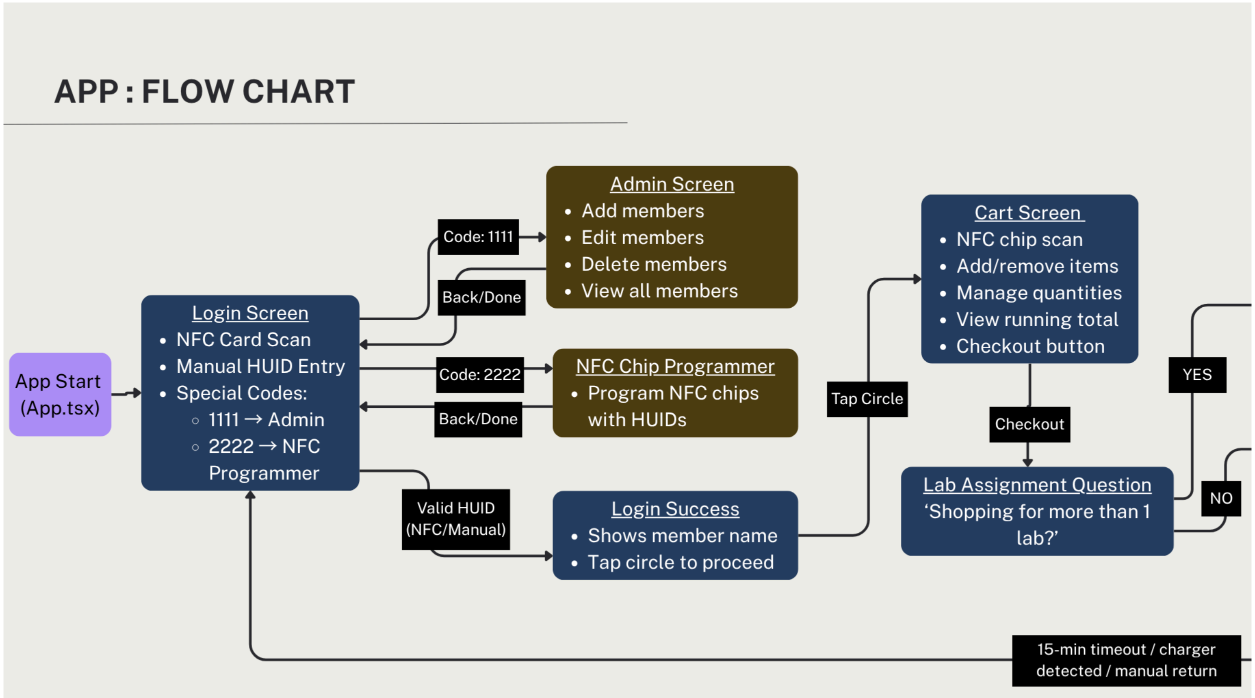
Table 3.2.2 — Build Environment Issues and Resolutions

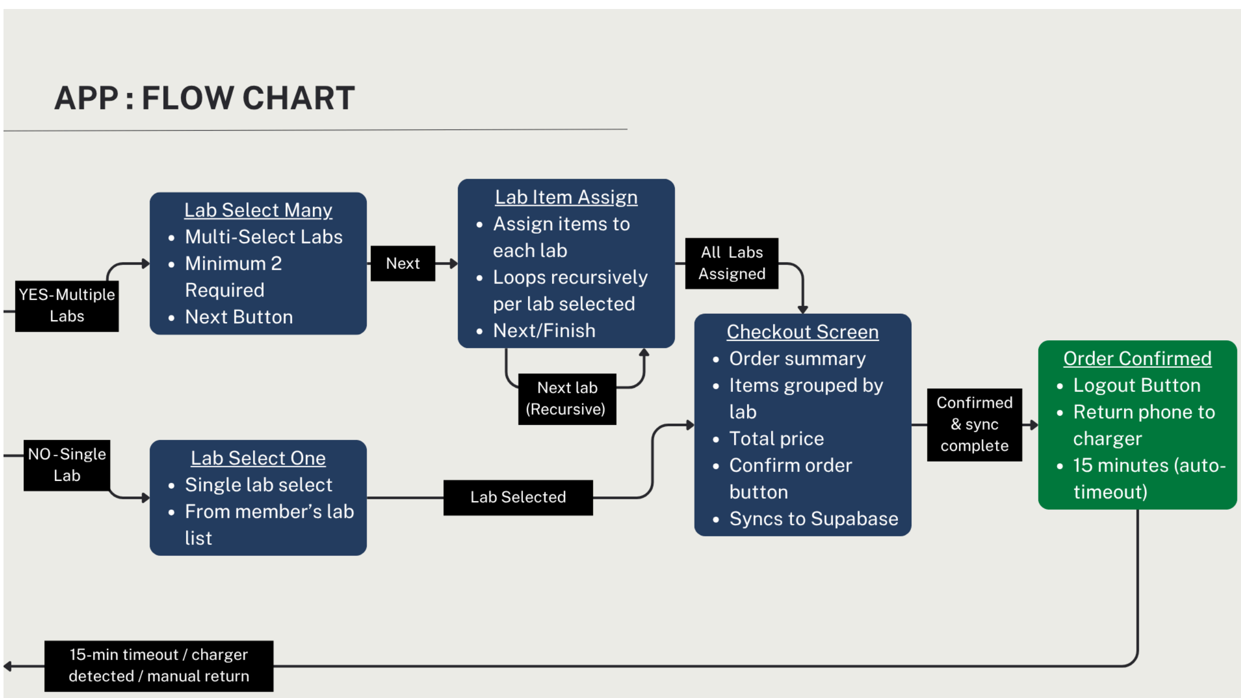
Issue Encountered	Resolution
32-bit JRE insufficient heap allocation	Configured org.gradle.java.home in gradle.properties to Android Studio's bundled 64-bit JBR
Android SDK path not found	Created android/local.properties with explicit sdk.dir
expo-barcode-scanner Kotlin compilation failure	Package removed (deprecated; not referenced in codebase)
NFC Tag Dispatch routing failure	Authored plugins/withNfc.js config plugin and rebuilt the native project

requestTechnology approach unreliable for MIFARE Classic cards	Migrated to registerTagEvent / DiscoverTag event-listener model

Of particular note is the NFC dispatch issue, which required the authoring of a bespoke Expo config plugin rather than a manual post-build modification, ensuring that the Android manifest changes are regenerated correctly on every subsequent expo prebuild invocation. Similarly, the migration from the technology-request API to the event-listener model for NFC scanning fundamentally changed the interaction paradigm: rather than actively acquiring and releasing a connection to a specific technology class, the hook passively listens for any tag event and queries the tag object for its identifier, making it robust against the diversity of MIFARE card variants encountered in practice.

3.2.8 Total App Flowchart





3.2.9 Summary

The front-end application was developed iteratively across five phases, progressing from a foundational member identity database through NFC integration, institutional branding, and a sophisticated multi-laboratory checkout flow. The resulting system is a self-contained native Android application capable of fully offline NFC-based authentication, real-time member enrolment and management, and flexible laboratory-attributed order submission. The architecture — a typed React Native codebase with a locally persisted data layer — requires no backend infrastructure, making it well suited to deployment as a standalone kiosk within a laboratory environment.

3.3 Back-end: Website Design

The project began with the creation of a Vite and React application scaffold, along with a GitHub Actions deployment pipeline. At this stage, the codebase remained unchanged from the default Vite template. The root component rendered the standard counter demonstration, and no application-specific logic had been added. This phase provided infrastructure groundwork rather than functional design.

The first meaningful design exploration introduced Supabase connectivity and established the earliest version of the application’s purpose. The interface was minimal, consisting of a basic heading, a raw HTML table with inline styling, and a hook that fetched rows directly from a stockroom items table. There was no routing, sidebar, authentication flow, or component structure. The application functioned primarily as a database connectivity proof of concept.

The next phase introduced Tailwind CSS, React Router, and an abstracted Supabase client module. This restructuring established the main layout pattern retained by the application: a sidebar navigation component paired with an authenticated main content area. The sidebar used a dark charcoal color scheme with indigo active-state accents. Access to protected routes was controlled through an authentication provider and login page. At this point, Items and Transactions were the only navigable pages. A mock data layer was also added so frontend development could continue independently of backend availability.

The visual identity was then revised to align with Harvard SEAS branding. The sidebar changed from dark gray to Harvard Crimson, and active navigation links adopted a white pill style with crimson text. The sidebar subtitle was updated to reflect the Harvard SEAS supply context. The Dashboard also became the default landing page, replacing the earlier redirect to the Items page. This dashboard displayed aggregate metrics such as total item counts, low-stock alerts, and recent transaction activity.

A later refinement phase expanded both the layout and data functionality. The Items page and global styles were improved, and the Dashboard was extended to include user management, lab assignment controls, and real-time Supabase subscriptions across multiple database tables. Two new pages were added: Inventory, for detailed stock-level management by stockroom, and Register, for account creation. NFC Tag ID support was also integrated into the data model. Routing was reorganized to separate public and protected routes more clearly, the sidebar was rendered conditionally based on authentication state, and a loading spinner was added during authentication resolution.

Overall, the design evolved through four main phases: initial scaffolding, backend connectivity prototyping, structural and branding development, and feature expansion. The final interface retained the Crimson sidebar, high-contrast active navigation style, authenticated two-column shell, Dashboard, Items, Transactions, Inventory, and Register pages.

The Dashboard page provides system administrators and stockroom managers a high-level overview of various aspects of the system, including recent transactions, total number of items, and low inventory. The Items page allows easy creation and deletion of unique items stored in the stockroom. The Transactions page provides a list of all transactions made with comprehensive metadata for inventory, billing, and auditing purposes. Finally, the Inventory page allows stockroom managers to update inventory according to their need.

3.4: Phone charging station

3.4.1: Version 1

When initially designing the charging station, our criteria were as follows: make a wall-mounted phone charging station that holds 3 phones and charges them wirelessly. The charging station must be visually appealing, easy to use, and robust enough for a lab setting with many users. After some initial market research, we found options that would meet some of these requirements, but not all. This led us to design our own phone charging station rather than purchasing a premade one. For instance, phone charging stations available on online retailers offered wireless charging but would require a shelf to sit on rather than mounting directly to the wall. We felt this would not be a professional solution, and would be hard to use since the chargers may fall off the shelf when being used heavily.

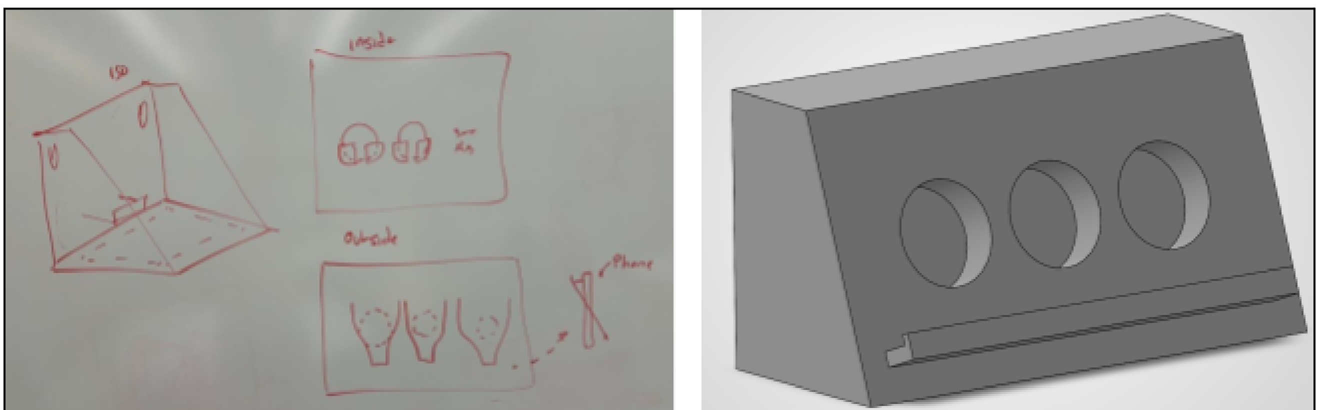


Figure 3.4.1: Initial sketches of phone chagring station (left). CAD model of initial phone charging station, showing overall shape, charger placement holes, and front lip for holding phones.

Our initial brainstorming consisted of drawing sketches on a whiteboard and proposing several different general designs. We considered having a stainless-steel wall-mounted plate from which the charging station would hang, and tabs built into the design sticking out to the sides for mounting. We also determined that the phone chargers would mount inside the box, with some internal structure to hold the end of the power cord and the block from which the chargers were powered. This would hide most of the electrical wiring, which would lead to a more polished design that was also less prone to unintentional user-caused damage. Ultimately, we settled on designing the charging station as one single 3D-printed part: three circular mounts cut into the design would frame each charger, held in place by a semicircular brace. Each charger would protrude slightly, and a straight lip along the front of the charger would sit all three phones in the charging position *(see figure 3.4.1)*. The charger would be essentially a rectangular prism, but with a slanted front face to ensure that the phones remained securely in the charger until a user removed one. We settled on the monopiece design because it would make manufacturing simple, since it would only require 3D printing in a single run *(see figure 3.4.1)*. 3D printing stood out initially as the top choice for manufacturing in part because of the variety of geometries needed to hold the circular chargers, rectangular phones, and charging block and extension cord. 3D printing would eliminate the need to use different manufacturing processes for each of these components of the design.

After settling on the monopiece 3D printed design, we started sourcing materials for the build so that we could refine the dimensions of the charging station and finalize the CAD to prepare for 3D printing. We settled on the Belkin 15W wireless charging pad, which was highly rated on Amazon, had a removable charger (which was an important factor for easy installation), and had a flat front as opposed to the rounded protrusions of some alternatives, such as the Amazon Basics option. The flat surface would help ensure a better interface between the phone and charger. We selected the Amoner USB-C charger block because it was highly rated and had

3 ports for charging, one for each phone. Finally, we selected a 15-foot interior extension cord from Grainger to ensure we could reach an outlet.

As we began to look into 3D printing options, we found that the 3D printers we have access to in the Harvard Active Learning Labs have a maximum build size of 14” x 10” x 14”, while our design has a maximum dimension of 16.5”. 16.5” was already the smallest this length could be, as constrained by the diameter of the chargers we specified for the build. Immediately, our idea was to divide the 3D print into three modular pieces, which we would print individually, then mount together to form the final product. However, after consulting with David Sekoll at the Harvard SEAS Machine Shop, we determined that such a geometry as we envisioned would be more successfully realized by the traditional manufacturing process of fingerlocked wood joinery. This process has long been used for making boxes and is known for its durability. Additionally, the manufacturing would be sped up by the use of a laser cutter for making the individual wood sheets of the box. Two wood thicknesses were immediately available – 1/8” and 1/4” – giving flexibility in how we designed components of the box that were formed from layers of wood glued together. This enabled complex geometries comparable to what a 3D printer would be able to achieve without the time-intensiveness, higher chance of print failure, resin cost, and questionable durability of 3D printed components. Thus, we decided to use a plywood finger-locked box design.

3.4.2: Version 2

Our second iteration of the mount introduced several key design changes from Version 1, particularly regarding structural integrity, charger accessibility, lid integration, and phone placement. After deciding to fabricate the product from wood, the team incorporated finger joints between the mount components to improve stability and overall structural rigidity.

One major design pivot involved the charger insertion method. In the initial concept, the charger was inserted through a slot in the back of the mount. However, due to the size of the charger and the need for easier access, the team decided to explore new options for charger placement. The two main designs that were discussed were inserting the charger through the front of the phone mount, and creating a layered assembly that would squeeze the charger in place between it and the mount. The team decided to eliminate the front inserting option due to the complexity of the geometry needed to properly support the phones once the charger was in place. The layered assembly was then fully developed and designed to mount directly onto the front face of the mount *(see figure 3.4.2)*. This new configuration would consist of 5 layers of laser cut plywood. The first two layers (starting from the front face of the mount) were designed to constrain the Belkin chargers. These consisted of two ⅛-inch plywood sheets designed to include force-fit holes, allowing the geometry to keep the chargers in place. The next three layers were made to fully support the phones against the chargers to ensure they would interface well *(see figure 3.4.2)*.

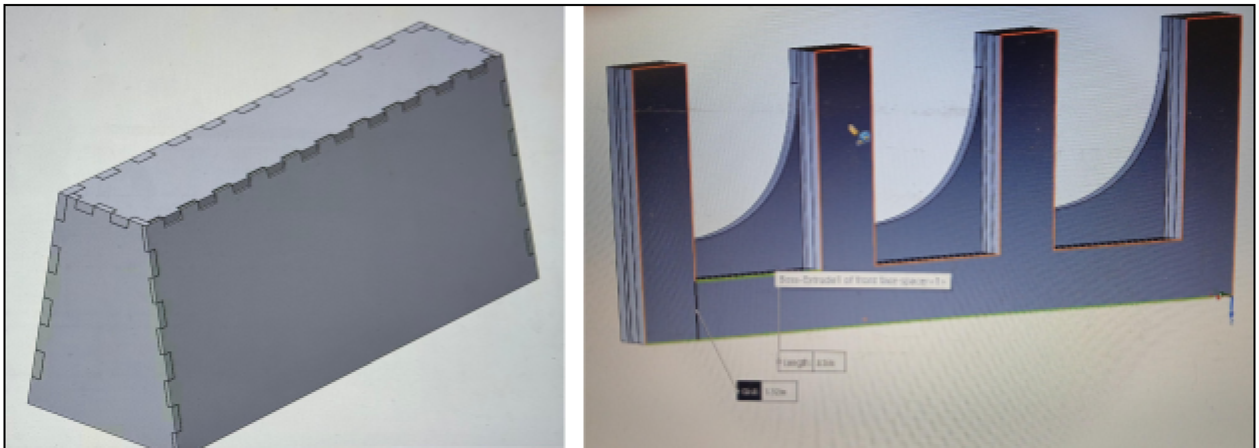


Figure 3.4.2: CAD of finger joints on solid mount (left). CAD of layered lid design (right).

Because the chargers needed to be inserted and enclosed within the layered structure, the team integrated the layered assembly into a lid mechanism. This lid was designed to hinge at the top, allowing the user to open the mount, place the chargers into their designated holes, and then close the lid over them. To evaluate this design, the team laser cut a cardboard prototype to visualize the assembly, assess the lid mechanism, and test whether the charger holes were positioned correctly for phone charging *(see figure 3.4.3)*.

Figure 3.4.3: Cardboard Prototype showing charger holes

Initial testing of the cardboard prototype showed that the phone placement did not reliably align with the chargers. After testing the charging setup outside of the box, the team lowered the phone position by ¾ inch to improve alignment. The team also found that charging was unreliable when the phone case was left on, regardless of the phone orientation. As a result, the design was modified to accommodate phones without cases, and the lid spacer dimensions were adjusted accordingly. After implementing these changes, the team moved forward with the first wood prototype, which incorporated the revised charger placement, lid mechanism, layered construction, and updated phone positioning *(see figure 3.4.4)*.

Figure 3.4.4: Completed V2 prototype with charging phone inside.

3.4.3: Version 3

Building on the Version 2 prototype, Version 3 focused on resolving shortcomings identified during testing and mock installation. The primary areas of improvement included cable management, lid fitment, latching, and overall structural integrity.

Following assembly of V2, a mock wall-mounted installation was conducted to evaluate cable routing and organization. While the rear cable tray successfully supported the primary wiring – three charger cables and an extension cord – excess cable length was visible beneath the unit. This introduced both aesthetic concerns and potential functional risks, such as snagging or accidental disconnection. To address this, excess wiring was first consolidated using zip ties. In addition, semi-permanent cable management mounts were installed on either side of the wiring path to provide improved strain relief and long-term stability. These additions complemented the existing cable tray and resulted in a cleaner, more secure wiring configuration.

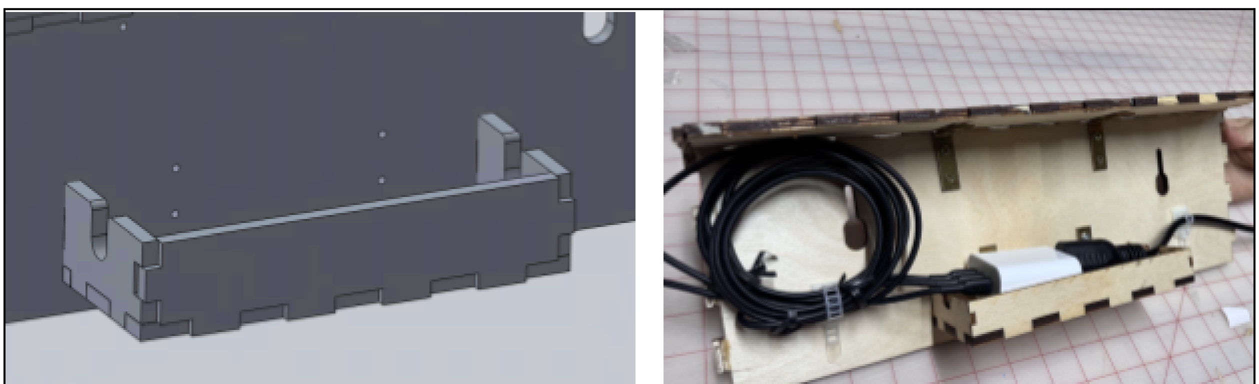


Figure 3.4.5: CAD model of internal wiring tray (left), Internal view of complete wiring (right)

During testing, incomplete lid closure was observed due to two primary interference points. The first occurred at the hinge, where the hinge screws protruded slightly into the closing path. Mitigation strategies included countersinking the screws as flush as possible and locally removing material from the wooden housing to accommodate the hinge geometry. While these adjustments improved closure, they did not fully eliminate interference.

The second, more critical, interference point arose from the interaction between the lid and the wireless chargers. The dual-layer lid design intentionally applies slight downward pressure onto the chargers, ensuring they remain flush with the phone contact surface to maintain reliable charging alignment. Because this feature was central to the design's functionality, it was preserved. As a result, an alternative method of securing the lid in the closed position was required – one that could tolerate minor hinge misalignment without compromising the chargers' positioning. This constraint directly motivated the integration of a latching mechanism.

An initial concept using embedded magnets was prototyped to achieve a flush, low-profile closure. However, the available magnets lacked sufficient holding force to overcome the hinge resistance and internal compression from the charger assembly. Consequently, this approach was deemed inadequate. The final design implemented a hook-and-eye latch system. This solution provided a simple, robust, and adjustable method of securing the lid, allowing fine-tuning of closure force through screw placement. Importantly, this design met the project requirement of ease of use without introducing unnecessary complexity such as keyed locking mechanisms, which were determined to be non-essential for the intended application.

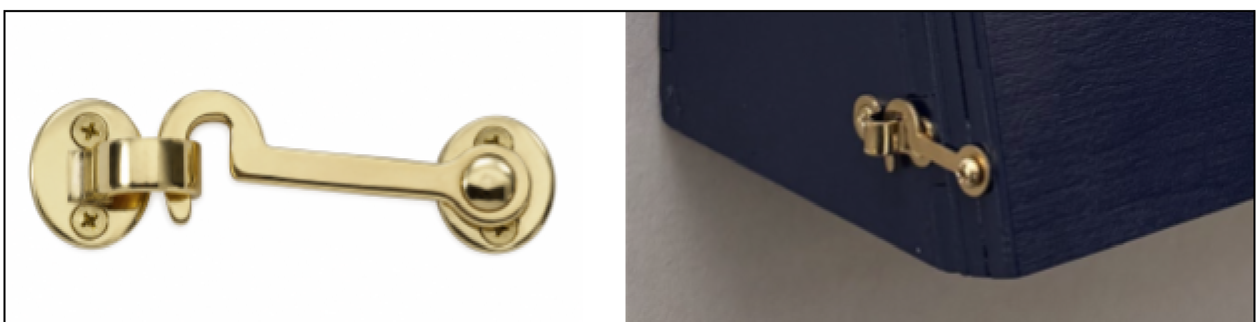


Figure 3.4.6: Hook and eye latch model (left), latch on the final design (right)

As the design approached finalization, additional consideration was given to durability under real-world use conditions. Analysis of the structure revealed potential failure points in regions where long, unsupported plywood spans could experience high bending moments from user interaction (like pressing or leaning). To reinforce these areas, aluminum L-brackets were installed along the internal perimeter of the enclosure. This reinforcement redistributed applied loads and significantly improved the rigidity and robustness of the assembly, reducing the likelihood of deformation or joint failure during operation.

Figure 3.4.7: L-bracket model (left), Internal placement of L-brackets and ties (right)

4 - Final Design

4.1 - Physical Design

After incorporating the improvements identified in V3, the team arrived at a finalized charging station design that meets both functional and aesthetic requirements. The final product features a plywood finger-jointed enclosure with an internal layered lid assembly that securely houses three wireless charging pads. The lid applies controlled compression onto the chargers to ensure consistent alignment between the charging coils and the phones, directly addressing the reliability issues observed in earlier prototypes. To accommodate minor geometric tolerances introduced by hinge interference while maintaining this compression, a hook-and-eye latch mechanism was integrated, allowing the lid to remain securely fastened without compromising charger positioning.

From a usability standpoint, the hinged lid enables easy access to internal components, allowing staff to easily install, replace, or service chargers and wiring. When opened, the lid remains stable in an upright position, exposing the internal layout for maintenance without requiring additional support. In the closed position, the latch ensures the enclosure remains secure during repeated daily use in a high-traffic lab environment.

The phone holding slots have been carefully tested to ensure that charging occurs regardless of the orientation in which the phone is replaced. This is critical for the functionality of the system; in order to remain operational at any given time, the phones must be charged properly, without requiring careful consideration and placement from the user. The positioning of the phones in the holsters guarantees perfect charging every time.

Figure 4.1.1: Final Design - Closed (left) and Opened (right)

The exterior of the unit was finished with multiple coats of paint, giving the device a clean, minimalistic, and professional appearance suitable for deployment within the Wyss Institute. The final design successfully balances manufacturability, robustness, and user experience, resulting in a polished, display-ready product that fulfills the original project criteria while addressing the practical constraints encountered throughout the prototyping process.

Figure 4.1.2: Final Design - Charging

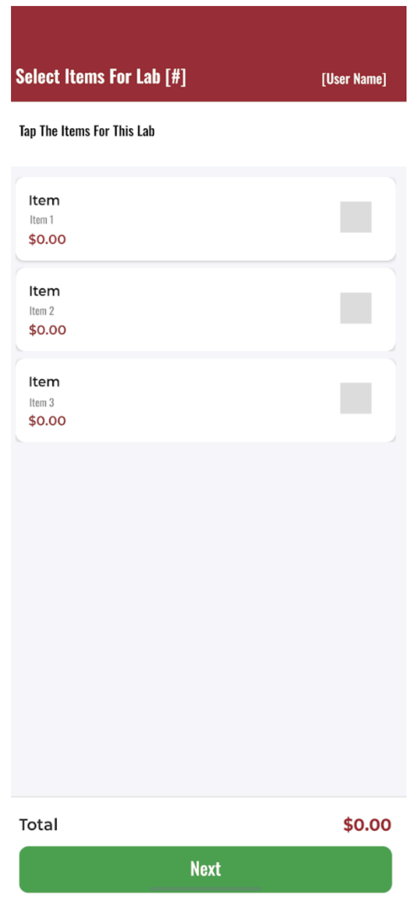
4.2 - Application

This application is designed to support a member-based checkout process using a phone-based interface. It allows members to log in through an NFC card scan (tap your physical HUID to the back of the phone) or manual HUID entry, add items to a cart, assign those items to one or more labs, review the order, and complete checkout. The application also includes administrative access and NFC programming functions that are available through special access codes from the login screen.

When the application is launched, the user is directed to the "Login Screen". From this screen, the user may authenticate by scanning their HUID or by manually entering their HUID. Once the application recognizes a valid HUID, the user is taken to the “Login Success” screen, where the member’s name is displayed. This confirmation step ensures that the correct user has been identified before proceeding. The user then taps the circle on the screen to continue into the shopping workflow.

The "Login Screen" also provides access to two restricted functions. Entering the code 1111 (temporary) opens the administrative section of the application, while entering the code 2222 (temporary) opens the NFC chip programming section. These access points are not part of the standard checkout workflow and should only be used by authorized personnel. The administrative section allows approved users to add members, edit member records, delete members, and view all members. Once the necessary administrative task has been completed, the user can return to the “Login Screen” by selecting Done.

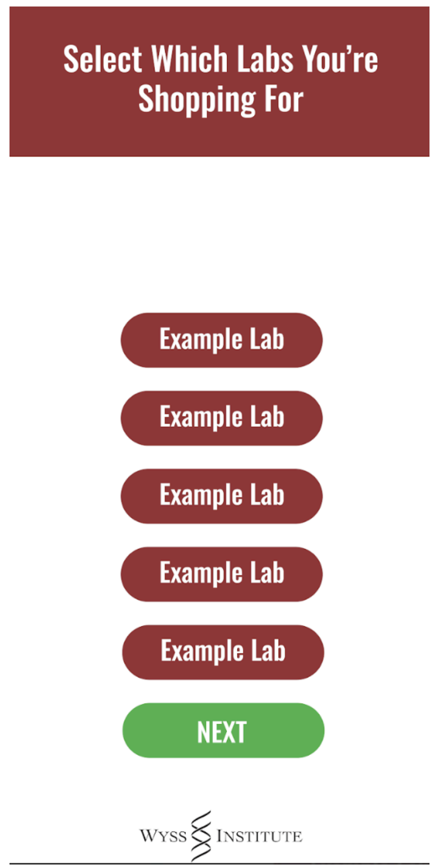
The NFC chip programming section is used to program NFC chips to specific items in the stockroom. This function is intended for situations where items in the stockroom are being issued, replaced, or updated and their corresponding NFC chips need to also be updated. Because the NFC chips all look identical, take special care not to mix up NFC chips. After programming is complete, the user can return to the "Login Screen" by selecting "Done."



After a standard user successfully logs in, the application displays the member's name and prompts the user to continue. Once the user taps the circle, confirming their identity, the application opens the "Cart Screen." The "Cart Screen" is the main shopping interface. From this screen, the user can scan item NFC chips, add items to the cart, remove items, adjust quantities, view the running total, and proceed to checkout. The user should confirm that each scanned item appears correctly and that all quantities are accurate before selecting the "Checkout" button.

After the user selects "Checkout," the application asks whether the order is for more than one lab. This question determines how the application handles lab assignments. If the order is for a single lab, the user is directed to the "Lab Select" screen. On this screen, the user selects one lab from the member's available lab list. Once the lab is selected, the application proceeds directly to the "Checkout Screen." This workflow should be used when all items in the cart are intended for the same lab.

If the order is for multiple labs, the user is directed to the “Lab Select - Multiple” screen. In this workflow, the user selects all labs that will receive items from the order. A minimum of two labs must be selected in order to continue. After selecting the applicable labs, the user proceeds to the “Item Assign” screen. The application then guides the user through assigning items to each selected lab. This assignment process repeats for each lab until all selected labs have been assigned their appropriate items. Once all lab assignments are complete and all items are accounted for, the application advances to the Checkout Screen.



The Checkout Screen provides the final review before submission. It displays the order summary, the items grouped by lab, the total price, and the option to confirm the order. The user should review this information carefully to ensure that the correct items, quantities, lab assignments, and total cost are shown. When the order is accurate, the user confirms the order. At that point, the application submits and synchronizes the completed order to Supabase.

Once synchronization is complete, the application displays the Order Confirmed screen. This screen indicates that the transaction has been completed successfully. The user may then select “Logout.” The application is also designed to logout automatically under specific conditions: inactivity for fifteen minutes or detection that it has been returned to the stand and is charging. This ensures that the application is ready for the next user and that member information is not left active on the device. At this point, the user should take the phone and return it to the charging station.

Overall, the application follows a structured process beginning with member authentication and ending with order confirmation. The primary user workflow consists of logging in, adding items to the cart, assigning the order to one or more labs, reviewing the checkout screen, and confirming the order. The administrative and NFC programming workflows are separate restricted functions and should only be accessed by authorized personnel. This separation helps preserve the integrity of member records, NFC credentials, and order data while supporting a clear and repeatable checkout process for regular users.

4.3 - Website

The final web interface serves as the secondary user-facing component of the stockroom management system. It is designed to support both a lab member workflow and a protected administrator dashboard, while remaining accessible through a standard web browser without additional software installation. This design minimizes technical barriers for lab members and provides administrators with a centralized interface for managing inventory, users, and stockroom activity.

The administrator side of the system is organized around a protected dashboard that requires valid login credentials. Once authenticated, administrators access a main interface. The sidebar contains four primary sections: Dashboard, Items, Transactions, and Inventory. The active page is visually highlighted to help users remain oriented within the system, while user identification and sign-out controls are positioned at the bottom of the sidebar. The main content area updates dynamically based on the selected section, and the system uses Supabase to store, retrieve, and synchronize data in real time.

The Dashboard acts as the main overview page for the stockroom system. It displays the total number of tracked items, the number of items below their minimum quantity thresholds, recent transaction activity, and a table of low-stock items. Low-stock entries include the item's current quantity, defined minimum quantity, and associated stockroom, allowing administrators to prioritize restocking without manually checking each storage location. The Dashboard also supports user and lab management by allowing administrators to view registered users and assign them to laboratory groups. These lab assignments make it possible to associate checkout activity with the correct research group for usage tracking and future cost accounting.

The Items page functions as the system's supply catalog. Each item record includes a display name, product code, category, and listed price. Administrators can search for items by name or code and filter them by category, such as Containers, Safety, Cleaning, and Miscellaneous. This organization mirrors the physical structure of the stockrooms and helps administrators locate supplies efficiently as the catalog grows. New items can be added through the item creation interface, while existing items can be removed from the catalog when no longer needed.

The Transactions page provides a complete audit log of inventory movement across the system. Each transaction records the date and time, user, stockroom, item name, quantity, and transaction type. Administrators can filter the log by user, stockroom, and transaction type, and transactions can be sorted by date. This page supports both operational review and accountability by showing which labs are using supplies, how quickly items are being consumed, and where discrepancies between logged transactions and physical inventory may have occurred. Transaction records may also include a notes field containing JSON metadata, such as lab member name and lab assignment, which provides a basis for more detailed lab-level reporting in future versions.

The Inventory page provides detailed stock management across multiple stockrooms. Stockrooms are displayed as collapsible sections, each containing a searchable table of stored items. For each item, administrators can view the product name, product code, NFC tag ID, current quantity, minimum quantity threshold, and status indicator. The status labels use color coding to show whether an item is acceptable, low, or out of stock. Quantity and minimum quantity values can be edited directly in the table, reducing the friction of routine inventory updates. NFC tag IDs can also be added or edited, linking the digital inventory record to the physical stockroom system. When an NFC-tagged item location is scanned with a compatible device, the system can identify the corresponding item and support checkout activity.

The final product gives lab members a browser-based path for checking out the availability of supplies, while giving administrators the tools needed to manage catalog items, monitor inventory, identify shortages, review transactions, assign users to labs, and maintain accurate records across multiple stockrooms. The interface emphasizes clear tabular displays, direct editing, color-coded status indicators, and real-time synchronization, allowing the system to support multiple administrators and a growing inventory within a unified platform.

5 - Future Considerations and Implementation

5.1 Manuals

5.1.1 - User Manual

Application Use

To enable new users to quickly learn how to use the StockSmart system, a user manual was developed based on the instructional video for users. This manual was developed as a 1-page, easy-to-read, guide, suitable for posting at the point of use to enable self-initiated training. In the implementation of the StockSmart system, this guide should be printed and laminated and affixed to the wall next to the phone mount.

StockSmart: User Guide

Entry

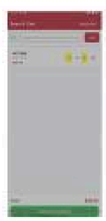
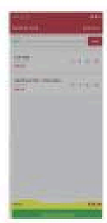
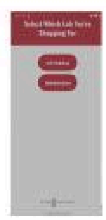
1. *Take a phone from the charging station located next to the door.*
2. *Turn on phone and tap Harvard issued ID on back to log in.*
3. *Tap phone on reader next to the door to access stockroom.*



1) 2) 3)

Item checkout

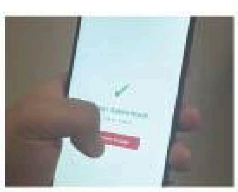
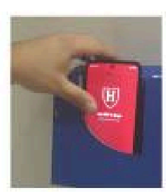
4. *Scan the NFC tag next to each item you take by holding the phone up to the tag.*
5. *Use plus and minus buttons on phone to adjust quantities as needed.*
6. *After adding all your items, click the checkout button at bottom of page.*
7. *Follow instructions on phone to indicate which labs you are shopping for.*
8. *After reviewing your order, tap confirm at bottom of page.*

4) 5) 6) 7)

Exit

9. *Exit the stockroom, then log out of phone by pressing the red button.*
10. *Place phone back in charging station and ensure that the phone is charging before leaving.*

9) 10)

Website Usage

Currently does not exist. In future iterations, users will be able to view available items in the catalog without editing capabilities, request item withdrawals or transfers, view their own transaction history. access stockroom-specific inventory views filtered to their assigned lab or location, and submit restock requests for low items.

5.1.2 - Administrator Manual

Application Side

The "Login Screen" in the application provides access to two restricted functions. Entering the code 1111 (temporary) opens the administrative section of the application, while entering the code 2222 (temporary) opens the NFC chip programming section. These access points are not part of the standard checkout workflow and should only be used by authorized personnel. The administrative section allows approved users to add members, edit member records, delete members, and view all members. Once the necessary administrative task has been completed, the user can return to the “Login Screen” by selecting Done.

The NFC chip programming section is used to program NFC chips to specific items in the stockroom. This function is intended for situations where items in the stockroom are being issued, replaced, or updated and their corresponding NFC chips need to also be updated. Because the NFC chips all look identical, take special care not to mix up NFC chips. After programming is complete, the user can return to the “Login Screen” by selecting “Done.”

<u>Website Usage</u>

All access to ES96 Admin begins at the Login page. Administrators enter their Harvard email address and password to access the system. After authentication, the Dashboard opens as the default landing page. The sidebar footer displays the current user's Harvard University Identification and includes a Sign Out button. New users may use the Register page to create an account, but administrator permissions must be properly provisioned in the backend.

The Dashboard provides a real-time overview of the stockroom system. It displays the total number of catalog items, current inventory records across all stockrooms, the five most recent transactions, and the registered user list. Items at or below their minimum quantity appear in the Low Stock section, allowing administrators to identify supplies that require restocking. The Dashboard also includes a lab assignment panel where administrators can assign users to Machine Shop, Living Materials, Soft Robotics, Biofabrication, Microfluidics, or Imaging. These assignments are saved directly to the database. Dashboard data updates automatically when inventory, transaction, or user records change elsewhere in the system.

The Items page is used to manage the supply catalog. Administrators can search items by name or code and filter the catalog by category. Each item record displays its name, unique code, assigned category, and unit price. To create a new catalog item, administrators select Add Item, complete the modal form, and save the entry. Items must be created in the catalog before they can be added to stockroom inventory.

The Inventory page is used to manage physical stock across Stockroom A, Stockroom B, and Stockroom C. Each stockroom appears as a collapsible section containing a table of stored items. Each row shows the item name, current quantity, minimum quantity threshold, status, and NFC Tag ID. The status badge indicates whether stock is Out, Low, or OK. Administrators can edit quantity and minimum quantity values directly in the table, and changes are saved to the database. Administrators can also use Add Inventory to assign an existing catalog item to a stockroom with an initial quantity and optional minimum threshold. NFC Tag ID fields may be viewed and edited for future physical tag integration.

The Transactions page provides a read-only audit trail of supply movement. Each record includes the item, quantity, transaction type, stockroom, user, and timestamp. Transactions are ordered with the most recent activity first. This page is used to verify recorded changes, review additions and withdrawals, support accountability, and investigate inventory discrepancies.

Administrators should sign out using the sidebar footer when finished. Because the system synchronizes database changes in real time, multiple administrators may work concurrently while viewing current inventory, transaction, and user information.

5.2 - Video Walkthrough

In order to implement this improved system successfully, the team recognized that technical functionality alone would not be sufficient. The new workflow introduces several unfamiliar steps for users, including phone-based authentication, NFC-enabled item scanning, and digital checkout tied to lab attribution. Because the stockroom is used frequently and often under time constraints, even small points of confusion could lead to incorrect usage, incomplete transactions, or resistance to adoption. As a result, it became essential to provide a clear and standardized method of instruction that would ensure consistency across all users.

To address this need, the team developed a short walkthrough video that demonstrates the complete user experience from entry to exit. The script of the video walks users through each step in detail. It begins with the entry process, where users are instructed to retrieve a phone from the charging station, power it on, and log in by tapping their Harvard ID on the device. The video then shows how the authenticated phone is used to unlock the stockroom door. This portion emphasizes both the physical interaction with the device and the importance of proper login for tracking user activity.

The next section of the script focuses on item checkout. Users are guided through selecting materials and scanning NFC tags associated with each item to add them to a digital cart. The script explains how to adjust quantities using the interface and how to scan additional items as needed. It also highlights the checkout process, where users must specify the lab or labs they are purchasing for before confirming the transaction. This step is critical because it directly supports the project's goal of accurate, item-level billing and proper cost attribution.

The final portion of the script covers the exit procedure. Users are instructed to log out of the phone after leaving the stockroom and to return it to the charging station, ensuring it is properly docked and ready for the next user. This step reinforces system sustainability and prevents issues related to device availability or user tracking.

A video format was chosen because it allows users to see both the physical and digital components of the system in action. Written instructions alone may not fully convey how to interact with the phone, where to position the ID for scanning, or how the NFC tagging process works in practice. By visually demonstrating each step, the video reduces ambiguity and helps users build confidence in using the system correctly. This is particularly important for a shared environment with many users of varying technical familiarity.

Additionally, the video serves as a scalable training tool that can be reused for onboarding new lab members and reinforcing correct behavior among existing users. Consistent usage of the system is directly tied to the success of the project, since accurate billing and inventory tracking depend on reliable data input from every transaction. Providing a clear, repeatable instructional resource helps ensure that the redesigned stockroom system achieves its intended impact.

Located below is a link to the video and its script, as well as various graphics from the video to help illustrate device usage:

Video Link: https://youtu.be/JquwvmRw6VQ

Script:

Entry:

To access the stockroom. First, take one of the phones from the charging station located next to the door.

After turning on the phone, tap your Harvard issued ID on the back of the phone to log in.

Once logged in, tap the phone on the reader next to the door to gain access to the stockroom.

Item Checkout:

After deciding which items you want, approach the box and scan the NFC tag to add the item to the cart.

You may use the plus and minus buttons to adjust the amount, and if you need to add a different item to the cart, scan its respective NFC tag.

After adding all your items, click the checkout button at the bottom of the page.

Make sure to indicate whether you are shopping for multiple labs and which lab or labs you're shopping for. After reviewing your order, tap confirm at the bottom of the page, and now you may exit the stockroom.

Exit:

After exiting the stockroom, first log out of the phone by pressing the red button. Then, place the phone back in the charging station and ensure that the phone is charging before leaving.

Figure 5.2.1: Entry via phone

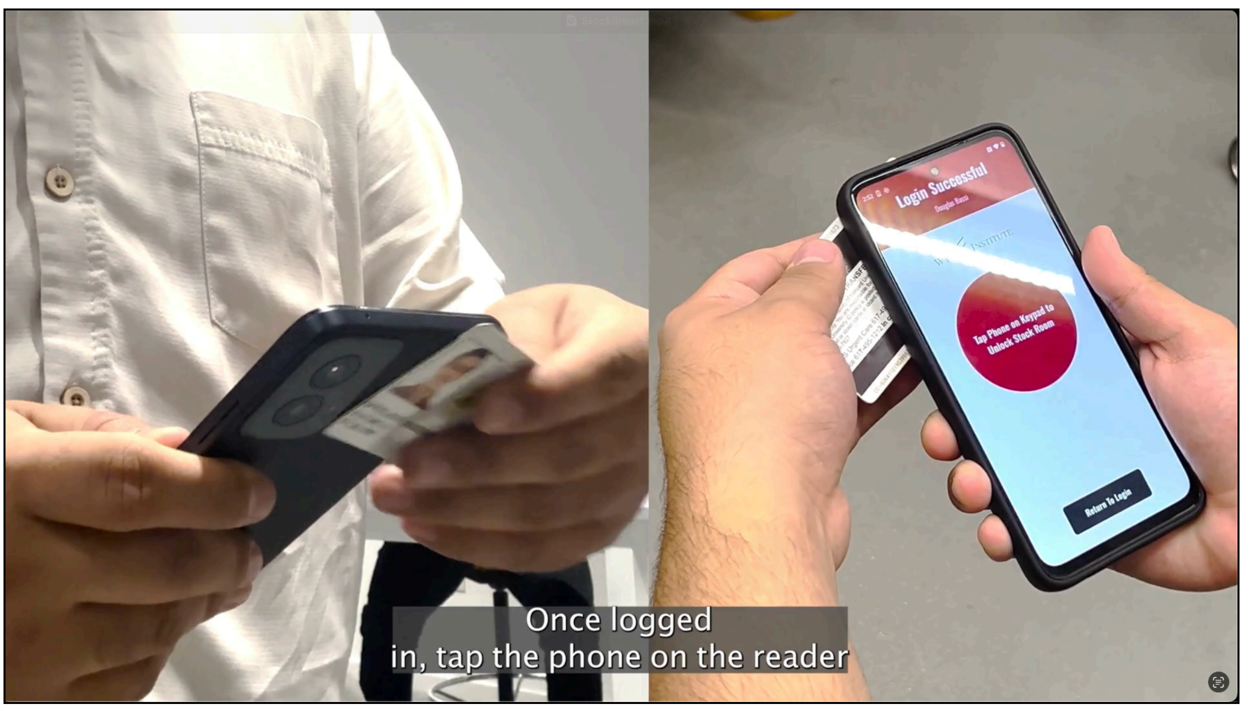


Figure 5.2.2: HUID recognition using phone reader

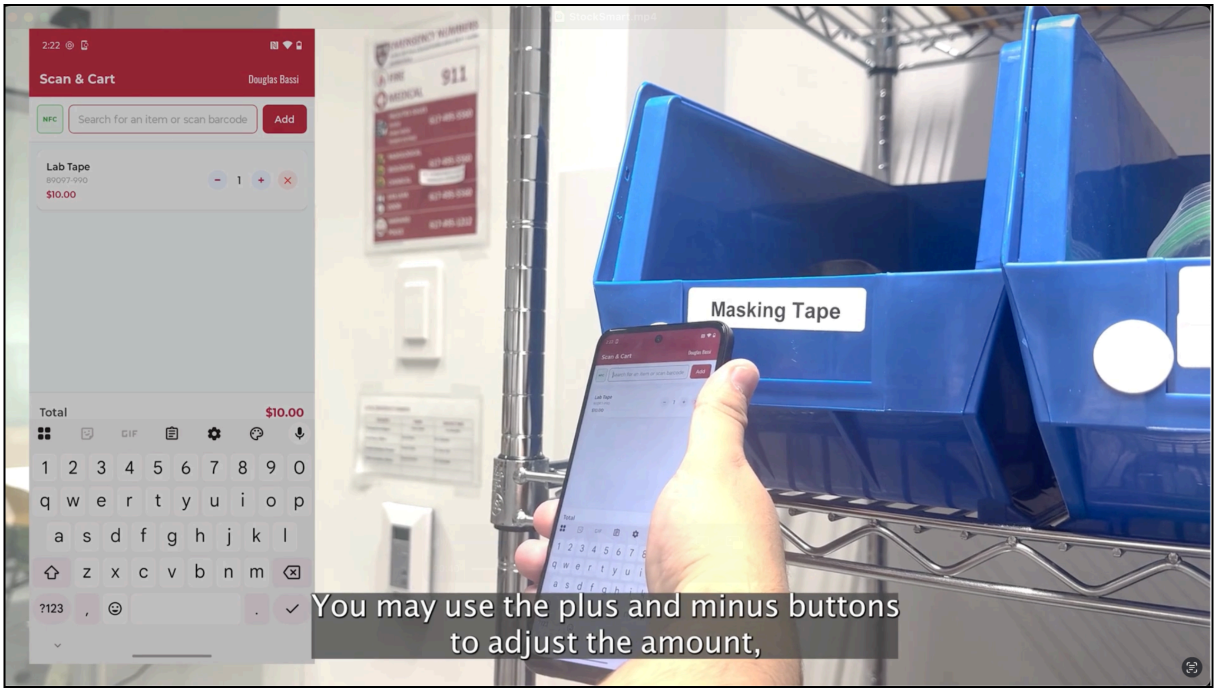


Figure 5.2.3: Adding items to the cart

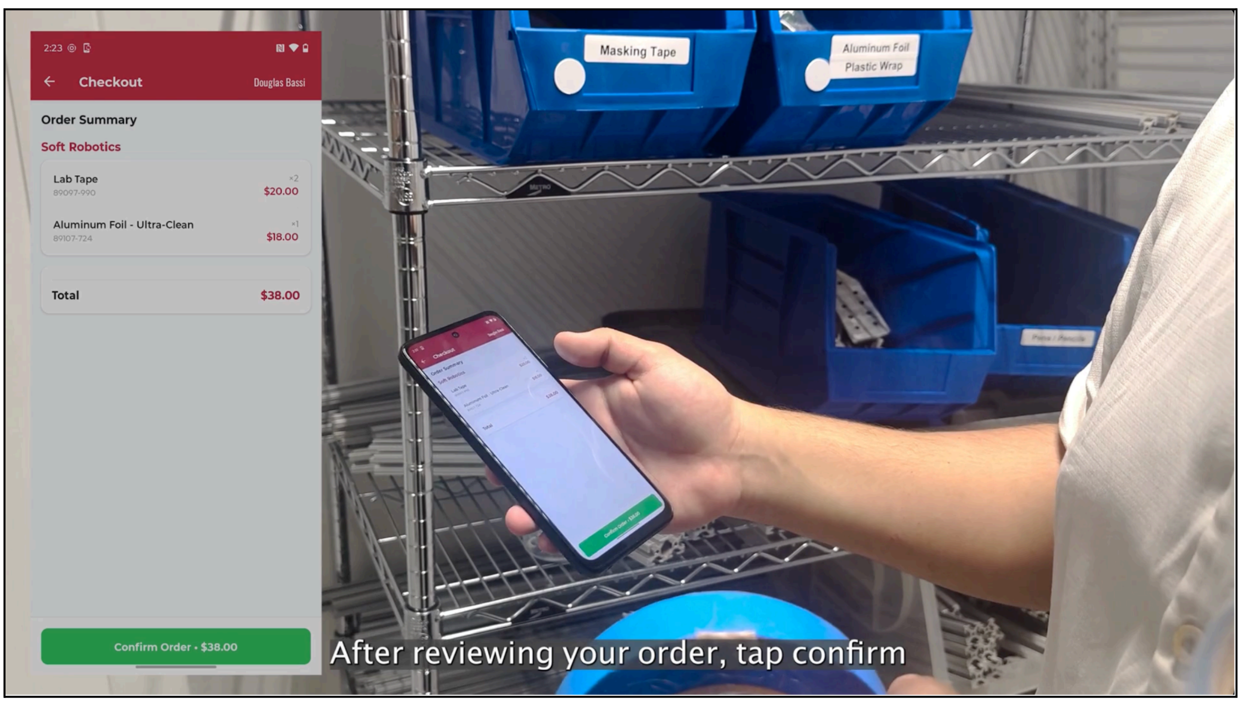


Figure 5.2.4: Ordering items using android application

6 - Conclusion

The StockSmart system represents the culmination of a semester-long engineering design process undertaken by the ES96 class in partnership with the Wyss Institute at Harvard University. Beginning with broad research across all five of the Institute's divisions, the class identified a clear, high-impact problem: the Wyss' shared stockrooms relied on a per-visit billing model that charged labs equally regardless of how much they consumed, producing financial inequity and operational inefficiency across the Institute.

Through a structured ideation and elimination process, the team evaluated seven candidate solutions and converged on a three-part NFC-based inventory management system consisting of a native Android mobile application, an administrative web dashboard, and a custom-built wall-mounted wireless charging station. The NFC solution was selected because it uniquely balanced user experience, accountability, adoptability, and implementation scope. This directly addressed the shortcomings of both the passive pressure-plate approach and the honor system dependent software-only alternative.

Prototyping proceeded across three parallel workstreams. The mobile application was developed through five iterative phases, delivering NFC-based HUID authentication, real-time item scanning, multi-laboratory cart assignment, and automated session management within a self-contained Android kiosk. The administrative website, built with Vite, React, and Supabase, provides stockroom managers with live dashboards, inventory tracking, transaction logging, and user management across multiple stockrooms. The charging station evolved through three hardware iterations, resolving challenges in wireless charging alignment, lid fitment, cable management, and structural integrity, ultimately yielding a durable, professionally finished plywood enclosure ready for lab deployment.

To support adoption and long-term usability, the team produced a one-page user quick-start guide, a comprehensive administrator manual, and a video walkthrough demonstrating the full checkout workflow from entry to exit. Together, these materials ensure that both regular users and system administrators can operate StockSmart with minimal onboarding friction.

While the current prototype meets the core functional requirements established at the outset of the project, several enhancements remain for future implementation. These include incorporating client feedback into a deployment-ready software build, expanding the user-facing website to support transaction history and restock requests, and completing final hardware installation at the Wyss. With these steps complete, StockSmart will provide the Wyss Institute with a fully automated, usage-accurate billing and inventory tracking system — eliminating longstanding financial inequity and reducing the operational downtime that has quietly eroded research efficiency across its shared laboratory infrastructure.

7 - Bibliography

Bloom, David. "Colossal Secures $200M to Accelerate De-Extinction and Genomic Innovation." Colossal Biosciences, January 16, 2025. https://colossal.com/colossal-secures-200m-to-accelerate-de-extinction-and-genomic-innovation/.

Gameto. "Gameto's $44M Series C to Help Redefine Reproductive Care, as First Patients Recruited for Pivotal Stem Cell Phase 3 Trial." PR Newswire, August 12, 2025. https://www.prnewswire.com/news-releases/gametos-44m-series-c-to-help-redefine-reproductive-care-as-first-patients-recruited-for-pivotal-stem-cell-phase-3-trial-302527828.html.

PitchBook. "Company Profile." Accessed May 1, 2026. https://pitchbook.com/profiles/company/460074-97.

Wyss Institute for Biologically Inspired Engineering. "Lactation Biologics: Increasing Milk Production for Healthier Babies." Harvard University. https://wyss.harvard.edu/technology/lactation-biologics-increasing-milk-production-for-healthier-babies.

Wyss Institute for Biologically Inspired Engineering. "OvaVasc: Enhanced Ovarian Tissue Engraftment for Fertility." Harvard University. https://wyss.harvard.edu/technology/ovavasc-enhanced-ovarian-tissue-engraftment-for-fertility/.

Wyss Institute for Biologically Inspired Engineering. "Wyss Institute Receives Wellcome Leap Funding to Develop First Human Organ Chip Model for Heavy Menstrual Bleeding." Harvard University. https://wyss.harvard.edu/news/wyss-institute-receives-wellcome-leap-funding-to-develop-first-human-organ-chip-model-for-heavy-menstrual-bleeding/.

Wyss Institute for Biologically Inspired Engineering. "Women's Health." Harvard University. https://wyss.harvard.edu/technologies/womens-health/.

Wyss Institute for Biologically Inspired Engineering. Harvard University. https://wyss.harvard.edu/.